Prepared for תיקון ליל שבועות at Harvard Hillel

בראשית / Genesis and physical cosmology

Benjamin V. Lehmann

Center for Theoretical Physics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

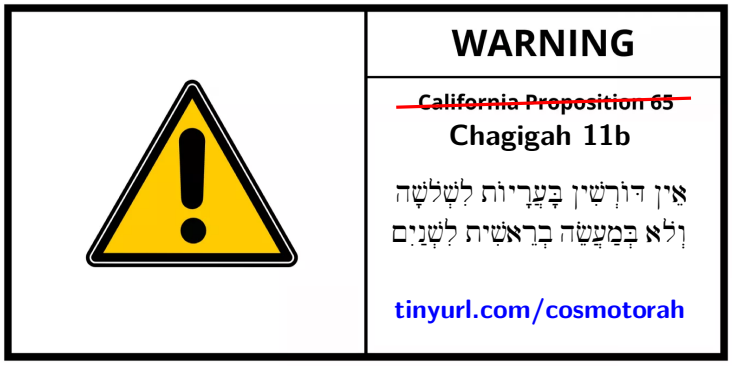
E-mail: benvlehmann@gmail.com

Date: June 10, 2024

Abstract: These notes are meant as a supplement to my shiur for תיקון ליל שבועות תשפ״ד (5784 / 2024) on reading the narrative of creation through the lens of physical cosmology. This shiur has been designed together with Jeff Cohen.

The first part of these notes focuses on the scientific aspects. I make some references to תנ״ך only to highlight interesting features that might invite further thought and learning. The second part of these notes focuses on the study of בראשית given the modern understanding of cosmic history. In this part, I explain why, as a scientist, I cannot accept a literal reading of the narrative. Still, I nonetheless present several ways of thinking about the relationship between these different cosmologies that may be interesting to readers of many different intellectual and theological backgrounds. My highest ambition is to engage with the text in a way that isn't intellectually lazy.

Since the scientific content of these notes is meant to provide additional details for the purpose of better understanding בראשית, they present the content in the form of an accessible retrospective summary. I make no effort to provide either historical context or mathematical rigor. As such, I have not included much discussion of *how* we know these conclusions to be reliable. I simply ask you to trust me—or to send me a message if you'd like to know more!



Contents

I Physical cosmology 2

1 Premises of physical cosmology: וַהֲחָכְמָה מֵאַיִן תִּמָּצֵא 2

2 The dynamical Universe: מִי שָׂם מָמַדֶּיהָ 4

3 Constituents of the Universe: עַל מָה אֲדָנֶיהָ הָטְבָּעוּ 7

4 The Big Bang: תֹּהוּ וָבֹהוּ 9

5 Quantum fields and inflation: וְחֹשֶׁךְ עַל פְּנֵי תְהוֹם 11

6 Phase transitions: וַיְהִי אוֹר 13

7 Hadronization and nucleosynthesis: וְרֹאשׁ עַפְרוֹת תֵּבֵל 19

8 Decoupling and recombination: וַיַּבְדֵּל אֱלֹקִים בֵּין הָאוֹר וּבֵין הַחֹשֶׁךְ 22

9 Structure formation: יְהִי רָקִיעַ בְּתוֹךְ הַמָּיִם 25

10 Stars, galaxies, and quasars: יְהֵי מְאֹרֹת בִּרְקִיעַ 27

11 Planet formation: יִקָּווּ הַמַּיִם מִתַּחַת הַשָּׁמַיִם אֶל־מָקוֹם אֶחָד 29

12 Anthropics and thermodynamics: לֹא תֹהוּ בְרָאָהּ לָשֶׁבֶת יְצָרָהּ 30

II Metaphysical cosmology 32

13 The literal reading (פשט) and its metaphoric extensions 32

14 The literary reading: creation in an ontological context 43

15 Mystical interpretations (קבלה) 49

16 Tripartite dialectical interpretation 52

17 Reconstruction: cosmological אגדה on בראשית א׳ 55

18 Final thoughts 58

Part I
Physical cosmology

1 Premises of physical cosmology: וְהַחָכְמָה מֵאַיִן תִּמָּצֵא

These notes discuss *cosmology.* Cosmology is the study of the Universe as a whole, making statements that apply on the largest scales. While cosmology includes the study of the history of the Universe, it is distinct from *cosmogony,* which concerns the origin of the Universe. The תנ״ך touches on both cosmology and cosmogony: it articulates a narrative of the origin of the material Universe, and it also makes certain statements about the contents and history of the Universe following its initial creation. As I will explain in these notes, modern cosmology has very little to offer in the way of cosmogony.

Today, cosmology is de facto identified with *physical cosmology,* which is the application of physics to the study of the Universe. Physical cosmology assumes that the same fundamental physical laws that we observe today also apply to the Universe at earlier times. Here, the word *fundamental* is doing a lot of heavy lifting: the conditions of the early Universe are so different, and the scales involved so much more vast, that many apparent physical laws of the late and local Universe simply do not apply—even what you might think of as bedrock principles, such as conservation of energy.[1] As such, it is worth taking a moment to understand how physics itself deals with the question of what is "fundamental."

One of the core principles of modern physics is that the effective laws of physics vary with *scale.* Observations at very different scales of length, time, or energy can yield drastically different results, and laws that look as though they are fundamental at one scale can instead emerge from deeper principles at other scales.

As I'm describing it now, this seems qualitative. For example, you might be thinking of the fact that calculations in Newtonian mechanics leave out corrections from relativity. These corrections are very small for objects that aren't moving quickly with respect to the speed of light, so the "deeper" theory of relativity apparently matches onto Newtonian mechanics, as long as you aren't driving well over the speed limit. This is true, but it isn't the whole story.

Consider, for instance, quantum chromodynamics (QCD). This is the theory of quarks and gluons, the constituent particles in protons and neutrons, which are themselves constituents of atomic nuclei. At very high energies, or short times, or short lengths—which turn out to all be the same thing, in a way—quarks behave as you would expect particles to behave. They're basically quantum billiard balls.

But zoom out, or wait longer, or do an experiment at lower energy, and you will find a remarkable result: a single quark *cannot exist* on its own. Any process that would give you a free quark involves enough energy to produce more quarks out of the vacuum, and the end result of these processes will *always* be pairs or triplets or higher numbers of quarks bound

[1] Yes, really. We just don't have space to get into it here.

together in states called *hadrons*. For many such processes, it doesn't even make quantitative sense to write the parameters of the theory in terms of quarks.

This is an example where it isn't sufficient to say that the quarks are "more fundamental" than the hadrons and call it a day. Rather, the physics is much better described by treating the hadrons themselves as though they were the fundamental objects. Accordingly, in this regime, instead of a theory of quarks, we write a theory of hadrons.

There is a rigorous, quantitative framework called *effective field theory* (EFT) that describes how different theories transform into one another as a function of scale. One of the basic results of EFT is that observations at one energy scale can be almost completely decoupled from physical phenomena with different intrinsic energy scales. QCD, for instance, has an intrinsic energy scale $\Lambda_{\text{QCD}} \simeq 100\,\text{MeV}$, or $10^{-11}\,\text{J}$. A physical process with energy $E \gtrsim \Lambda_{\text{QCD}}$ behaves as though the quarks are free particles, knowing nothing about hadrons, and a process with energy $E \lesssim \Lambda_{\text{QCD}}$ behaves as though the hadrons are the fundamental objects, knowing nothing about quarks.

Returning to cosmology, the significance of effective field theory is that *some cosmological phenomena take place on scales that we cannot probe by any other means.* In these regimes, our understanding of cosmology ends up informing our picture of fundamental physics, rather than the other way around. However, many important cosmological phenomena take place on scales that we *can* probe in many other ways. These regimes of overlap between different experiments and observables give us a way to test the idea that we can apply our knowledge of physics to faraway places and times in cosmic history. At the risk of spoiling the suspense, modern physics passes this test with flying colors, with spectacular consistency in the laws of that we observe now and billions of years ago, here on Earth and half a Universe away.

So, let us take the premise that we can write physical theories that model the dynamics of the Universe as a whole. We don't need to make any statement about what physics we consider to be fundamental—only that *the same physics should apply on the same scales.* Once we adopt this notion, the goal of cosmology becomes the same as that of any other scientific field: to develop *simple, falsifiable models.* While these terms mean the same as they do in other fields, let's take a moment to understand the implications.

If a model is simple but remains predictive, that generally means that it has explanatory power: there is some simple set of axioms and logical statements that give rise to observed phenomena. A model is *falsifiable* if those predictions are specific enough that we could conceivably obtain different experimental results and determine that the model is incorrect. This might seem roundabout, but remember, one can never prove that a model is *correct*—it could always fail in some unanticipated situation. On the other hand, if a model doesn't make any prediction specific enough that we could ever call it wrong, then it's vacuous—it has no nontrivial empirical meaning!

Modern models of cosmology have become extremely predictive and precise. While some topics of study in cosmology remain conjectural, beyond our current experimental capabilities, our modern understanding of physics is sufficient to describe an enormous range of cosmological phenomena. These physical models tell an extraordinary story of cosmic history.

2 The dynamical Universe: מִי שָׂם מָמַדֵּיהָ

To understand cosmology, we first need to understand the basic objects of cosmology. The first and most important player is the *spacetime* of the Universe.

You may understand space as separate from time, and as a static stage upon which matter plays out the laws of physics. If so, I must disabuse you of both of these notions.

1. In the theory of relativity, space and time are the same basic "substance." We say this because they can be interconverted while respecting the symmetries of the Universe. We would not say that distances measured ahead and distances measured to the right are "made" of different stuff, and we know that because we can change the meaning of "ahead" and "right" by simply rotating. The Universe is rotationally symmetric—it does not prefer any particular person's "ahead" direction—so we are entitled to measure by combining anyone's "ahead" and "right" distances, no matter their orientation.

   It turns out that the same is true, almost word for word, between space and time distances. There is a transformation analogous to rotation[2] that interchanges space and time, in the same way that rotation interchanges different directions. Since the Universe is symmetric under this transformation, measurements of time or space can be different between different observers. They are really just ways of labeling the same distance in a combined *spacetime*.

2. This spacetime is not static—far from it. Spacetime bends and curves in response to energy, and has *dynamics,* not unlike a rubber sheet. This directly influences the motion of objects on the spacetime. In fact, in the general theory of relativity, the force of gravity is nothing but the effect of objects following locally-straight trajectories through curved spacetime.

   If you ask a free-floating astronaut in space how she is moving, she could fairly respond that she is not moving at all in space. But even from that perspective, she is still moving forward in time, and therefore moving in spacetime. The curvature of spacetime is such that even if she only ever perceives herself to be moving forward in time, her distance to other objects will change.

   For an analogy, consider two friends, Alice and Bob, standing a thousand miles apart on Earth's equator. They each walk due North (forward in "time"). Alice reasons that since their direction of travel is parallel, they will remain a thousand miles apart. Bob agrees: he isn't moving East or West (i.e., in "space") at all. But since Earth's surface is curved, parallel lines don't stay parallel, and they meet each other at the North pole! In the same way, gravity can pull two astronauts together even when each insists that they are not accelerating, but simply moving forward in time.

What does this mean for cosmology? It means that in order to understand cosmic dynamics, we should start by understanding the dynamics of the cosmic spacetime itself.

[2] It's actually a hyperbolic rotation in the space and time axes.

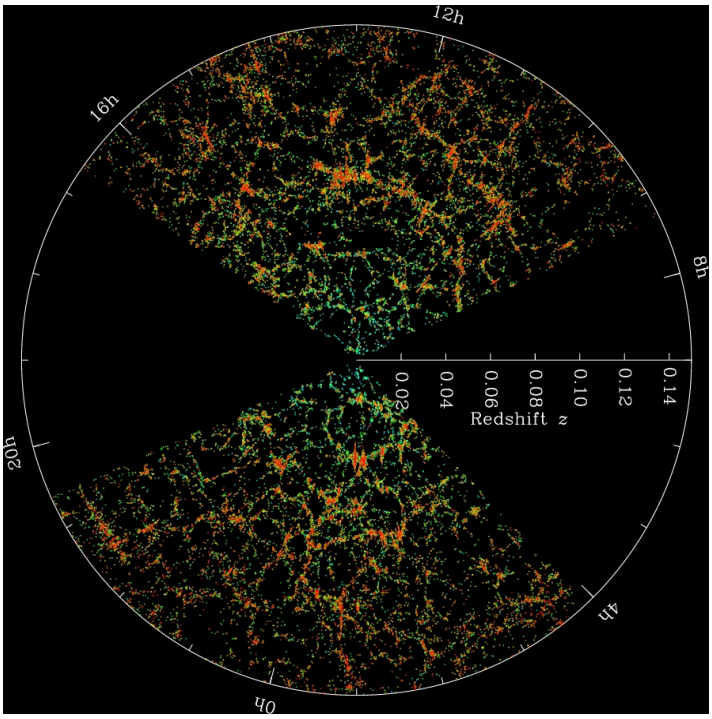


Figure 1. A map of galaxies in the Sloan Digital Sky Survey [1]. On the largest scales, these can be treated like a uniform gas, or "fluid."

In general relativity, the mathematical theory of spacetime, it turns out that we can say a lot about the spacetime of the Universe as a whole.[3] In particular, let's assume that the Universe is homogeneous and isotropic—that is, it looks the same from every point and in every direction—when averaging on very large scales. This is of course *not* true on small scales. Earth is very different from the surrounding Solar System, and even our galaxy is very different from the surrounding intergalactic medium! But when considering quantities that are averaged over enormous spatial volumes, containing thousands or millions of galaxies, the Universe *does* appear quite uniform, as shown in fig. 1. As long as we are only interested in making statements about the Universe on these largest scales, we are entitled to treat the spacetime as uniform. This immediately tells us that there are only a few different classes of spacetimes that can match our Universe while satisfying the equations of general relativity.

There are three basic shapes (*topologies*) of spacetimes that could behave like our Universe. One is a "flat" Universe, where the spatial part of the Universe's spacetime isn't curved. This is the simplest thing to imagine—an infinite, featureless space. But there are two other options. The spatial part of the Universe could be *positively* curved, meaning that it would look like a higher-dimensional version of a sphere. Just like the surface of the Earth, such a Universe would be finite, but it wouldn't have a boundary. If you traveled for eternity in

[3]Incidentally, general relativity was first formulated by Einstein, and was labeled "Jewish physics" by antisemites. That would have been a delightfully appropriate title for these notes.

one direction, you would eventually return to your starting point. It could also be *negatively* curved. This is a little harder to visualize. The spatial part of the Universe would locally look saddle-like, and the Universe as a whole would still be infinite.

The equations of general relativity further tell us that which one of these three possibilities is realized in nature depends on the energy content of the Universe. A Universe with high energy density is positively curved, and a Universe with low energy density is negatively curved. In order to be flat, the energy density of the Universe would need to sit right on the knife edge between positive and negative curvature. Interestingly, our Universe seems to do exactly that! We will revisit this point later on.

In each of these cases, general relativity also dictates how the Universe evolves with time. For each of the three possible shapes, there is only one parameter that can evolve: the overall "size" of the Universe. Now, "size" is a dangerous word: how can you ascribe a changing size to an infinite Universe? An infinite Universe can't "expand" in the sense of becoming larger. However, it *can* expand in the sense that distances between any two points increases. Imagine an infinite sea of rising raisin bread dough: even though the infinite loaf can't get bigger, the distance between any two raisins still increases. The technical term for this is *metric expansion,* and we measure the "size" in terms of the *scale factor,* a, which is set to be 1 at the present time. If the Universe is expanding, and was smaller in the past, then $a < 1$ in the past and $a > 1$ in the future.

The expansion or contraction of the Universe depends not only on the energy density of the Universe, but also on what kinds of energy are present. We'll get more specific in the next section, but roughly, different forms of energy, such as matter and radiation, exert different amounts of *pressure* on the Universe. Remember, we are thinking of the Universe as homogeneous, which means that the energy spread out through the Universe can be treated as a fluid. The behavior of a fluid is crucially connected to fluid pressure, and so is the Universe as a whole! We summarize the behavior of fluids of different forms of energy by relating their energy density, ρ, to their pressure, p:

$$p = w\rho. \tag{2.1}$$

Here w is called the *equation of state parameter.* For a "perfect fluid," like the large-scale constituents of our Universe, w is a constant number. For example, a fluid of dust particles is pressureless—no matter what ρ is, p is always zero. Such a fluid is completely characterized by the equation $w = 0$. A real Universe is populated by a combination of different fluids with different values of w. Given this data, we can specify the future evolution of the Universe—and we can also run the clock backwards to understand the history of the Universe.

Now let's turn to the fluids that are present in our own Universe.

3 Constituents of the Universe: עַל מָה אֲדָנֶיהָ הָטְבָּעוּ

Our Universe is well described by a combination of three fluids: matter, radiation, and dark energy. Let's take a moment to understand what this means.

First, what is matter, as opposed to radiation? Here, "matter" and "radiation" are technical terms—their definition in cosmology does not necessarily correspond to definitions in other fields. Each of these components corresponds to a fluid with a particular equation of state parameter w. In cosmology, *matter* means a pressureless fluid: $p = 0$, or equivalently, $w = 0$. *Radiation* means a fluid whose pressure is one third of its energy density: $p = \rho/3$, or $w = 1/3$. *Dark energy*, or a *cosmological constant,* is a fluid with $w = -1$, i.e., with $p = -\rho$. The expansion history of the Universe is determined by the relative amount of each of these components.

Notice that "radiation," as the term is used in cosmology, doesn't necessarily match what you think of intuitively as radiation. Really, any fluid in which the particles are relativistic behaves like radiation cosmologically. This means that electrons, for example, are part of the *radiation* component of the Universe when the Universe is hot enough. When the Universe cools, and the electrons are no longer relativistic, then they become part of the *matter* component.

The cosmological matter fluid also includes a less familiar form of energy that does not behave in the same ways as ordinary matter. This is *dark matter.* Dark matter interacts with other forms of energy gravitationally, but we don't yet know whether it interacts in any other ways. In particular, this means that dark matter is invisible. We have yet to detect dark matter particles by nongravitational means, but that doesn't mean it isn't important. Dark matter accounts for about 80% of all matter in the Universe today, and it plays a crucial role in creating the conditions for structures like stars and galaxies to form. We'll come back to dark matter in section 9. For now, the only thing to know is that dark matter is called "matter" because of the fact that it behaves like a pressureless fluid, with $w = 0$, and it dominates the matter fluid in our Universe.

The relationship between energy density and pressure in a fluid also influences how the energy density changes as the Universe expands. You would expect that as the Universe expands, the energy density of a fluid goes down. This is indeed the case for matter and radiation, but not at the same rate. Matter dilutes intuitively: if a patch of the Universe doubles in volume, spreading out the same matter over twice the volume, the matter density goes down by a factor of two. But the density of radiation does *not* behave this way. If the volume doubles, the density of radiation goes down by a factor of $2^{4/3} \approx 2.52$. In general, if the volume doubles, the energy density of a fluid scales by a factor of $2^{-(1+w)}$. This means that in an expanding Universe, fluids with larger values of w dilute faster, and eventually become subdominant to fluids with smaller values of w.

If you're paying attention to the algebra, you may have noticed something interesting: *dark energy doesn't dilute at all!* Since $w = -1$, when the volume doubles, the energy density scales by $2^{-(1-1)} = 1$. It doesn't change. (This is why dark energy is also called a

"cosmological constant.") So, what is dark energy? Conceptually, dark energy is the energy density associated with space itself. Remember, space is dynamical, not a static stage, and as such, it can have energy. Since this energy is an intrinsic property of space, it doesn't dilute as the Universe expands. Instead, as the Universe expands and makes more space, that new space brings with it new energy.[4]

You may have noticed that if dark energy doesn't dilute as the Universe expands, but other forms of energy do, then at some point in the future, we should expect dark energy to be the dominant form of energy in the Universe. In fact, it's not in the future—it's now. We happen to live at a moment in cosmological history where dark energy is surpassing matter as the dominant component of the Universe. That dark energy is driving further expansion, which speeds up the process. A Universe dominated by dark energy expands exponentially!

This gives us some clue about the future of our Universe. But we can also apply this reasoning in reverse, and make inferences about the past composition of the Universe. I've already alluded to the fact that the Universe is expanding—this is directly measurable. In the past, then, distances were shorter, and the Universe was *denser.* As we go back in time, the relative energy density of matter increases, dominating over dark energy. But as we go yet further back, the energy density of radiation increases sharply. At some point, over ten billion years ago, it dominated over the matter, and accounted for almost all energy in the Universe.

If we carry this exercise further, we can conclude that the early Universe was dense, radiation-dominated, and *hot.* Temperature increases as the density goes up, and if we go back far enough, the Universe was essentially a giant fireball. If we go back far enough, the entire Universe was hotter and denser than any medium we have ever created in the laboratory or seen in space. It was so hot that the particles that make up matter today would have been moving with relativistic speeds, and thus would have contributed to the *radiation* density instead of to the matter density. It was so dense that particles of all kinds couldn't move in straight lines without being almost immediately scattered.

Interestingly, we can't keep going back in time like this forever. At some point in the finite past, the scale of the Universe drops to *zero*—and this is where physical cosmology runs headlong into cosmogony.

[4]When I said that energy isn't conserved in cosmology, I wasn't kidding!

4 The Big Bang: תֹהוּ וָבֹהוּ

Now we come to the topic that generates the most breathless confusion in scientific discussions of the בראשית narrative. In this section, I hope to teach you a few things about the early Universe, and simultaneously set the record straight on some common misconceptions.

The Universe is expanding today, and we can precisely model the expansion in relation to the amount of energy of various kinds present. We can indeed run the clock back, and determine with great confidence that the Universe was denser and hotter in the past. It is certainly appropriate to ask about the end result of this backward evolution. If we go back in time as far as we can go, how did the Universe begin?

When we ask this question quantitatively, we find a strange result. At some time in the finite past, the scale factor drops to zero. That is, the mathematical result implies that all distances between all points in the Universe were zero. In other words, the theory predicts that the Universe itself collapses to a single point—a *singularity.* We might worry that this is an artifact of the exact properties we have assumed for our Universe. However, it turns out to be a much more general feature of cosmological spacetimes. In general relativity, any spacetime in which space has expanded on average over time *must* have such a singularity at a finite time in the past.

The naïve interpretation of this mathematical result is that we have discovered something about the origin of the Universe: that time had a beginning, $t = 0$, and that all of the energy in the Universe collapsed to a single point at that special time. This has apparent philosophical resonance. Does this singularity offer a replacement for the creator *ex nihilo*? Does it bolster a creationist viewpoint in which time is finite? Either way, does this pointlike origin of the Universe hold the key to understanding where we came from?

Now, I hate to disappoint, but the answer to each of these questions is *no.* The reason is simple: the singularity itself signals the breakdown of our mathematical framework for studying cosmology. All other considerations aside, we know that at the very highest energy densities, general relativity is incomplete. General relativity is a *classical* theory: it does not incorporate quantum mechanical corrections. These quantum corrections are small for almost all gravitational phenomena, but they would emphatically not be small for a singularity! This means that the entire theoretical framework is just not a valid tool for describing the earliest moments of the Universe. Was there really something like a singularity in the past? I don't know—nobody does.

You might think that this process of a pointlike singularity exploding into the full Universe is what we call the "Big Bang." It's not. The notion of a singularity has nothing to do with the Big Bang. The idea of the Big Bang is on much firmer footing, even though there is ample confusion about what the term means, even among professionals. So let's now let go of the idea of a singular physical cosmogony, and turn instead to the fabulously well-tested Big Bang theory. What is it, anyway?

The notion of a "Big Bang" begins with the observation that at some point in the far past, the Universe was both dense and hot. You might think that it is trivial that the Universe was

hot: after all, if you compress a gas, it heats up. If you compress enough gas forcefully enough, you can even ignite a star. But "hot" implies something deeply nontrivial: the Universe had a *temperature*. That is, the constituents of the Universe were in thermal equilibrium.

This is a strong statement. It means that to understand the state of the Universe at early times, we do not need to guess about the complicated distribution of energies amongst different particles. The particles were exchanging energy quickly enough that they maintained an equilibrium configuration, a state that yields unique predictions for essentially every macroscopic quantity. Thermal equilibrium implies relationships between energy density, temperature, the number of particles, the pressure, the distribution of different particle types, and so much more. It also means that it is nearly impossible to say what came before. The Universe could have started off with many different kinds of configuration before coming into a state of thermal equilibrium, but since thermal equilibrium always looks the same, that information is effectively hidden from us—although we'll shortly discuss some ways of getting at it nonetheless.

So, there is an enormously consequential step in cosmic evolution that drives much of what we see (and don't see) in cosmology today: the formation of a *thermal bath* out of whatever came before. This formation of a thermal bath is the Big Bang. Even without knowing exactly what happened to form this thermal bath, we know that it did form, and we can observe it quite directly. The subsequent evolution of the Universe was determined by the thermal properties of this dense fireball. That's the Big Bang theory, in a nutshell.

The attentive reader will notice that I alluded to תֹּהוּ וָבֹהוּ in the title of this section. Let's recall the description from בראשית, which reads:

וְהָאָ֗רֶץ הָיְתָ֥ה תֹ֙הוּ֙ וָבֹ֔הוּ וְחֹ֖שֶׁךְ עַל־פְּנֵ֣י תְה֑וֹם וְר֣וּחַ אֱלֹקִ֔ים מְרַחֶ֖פֶת עַל־פְּנֵ֥י הַמָּֽיִם׃

I do not subscribe to the belief that the creation narrative in תנ״ך can be mapped onto the chronology of physical cosmology. (In particular, I will make no apologia for apparent presence of מים at this stage.) However, for those who wish to indulge in the game of correspondence, I would suggest that the primordial state of chaos implied by תֹּהוּ וָבֹהוּ is not unlike the Big Bang fireball. In particular, I point out that this state of chaos does not mark the moment at which time originates. It is simply the unstructured, undifferentiated state of the Universe before its subsequent evolution, as with the Big Bang plasma.

In order to say anything about the details of the "Bang" itself, we need to have some notion of what came before the Big Bang. This is *not* a meaningless question—it's not like asking what came before time. The Big Bang, in this picture, happened *after* the origin of the spacetime of the Universe. As it happens, we actually have some good guesses about what came before, between the origin of time and the formation of a thermal bath. I'm going to tell you about the most popular idea: inflation.

5 Quantum fields and inflation: וְחֹשֶׁךְ עַל פְּנֵי תְהוֹם

The theory of inflation posits that the Universe underwent a period of exponential expansion just before the Big Bang. This solves a number of technical problems in cosmology that are otherwise difficult to understand.

In particular, it explains why far-separated parts of the Universe look statistically alike even though they haven't had enough time to exchange signals over the history of the Universe (the *horizon problem*). As a bonus, this period of rapid expansion also explains why the Universe must look very nearly spatially flat, appearing to be right on the knife edge between positive and negative curvature (the *flatness problem*). If the Universe expanded dramatically, then the radius of curvature is simply much larger than our line of sight, and the Universe looks flat for the same reason that Earth looks flat if you can't see very far ahead.

Now I want to explain what the Universe would have been like during the inflationary epoch, and how it would have led to the Big Bang. But to do that, I first need to explain some other concepts in physics. The Universe couldn't simply inflate of its own accord. As we have already discussed, the expansion of the Universe is directly related to what forms of energy it contains and in what amounts. You may have noticed that the exponential expansion that is characteristic of inflation is also characteristic of a Universe dominated by dark energy—so, could dark energy drive inflation? Well, not the dark energy that we know of today. Dark energy has a constant energy density, and the dark energy we observe in the late Universe has an extremely low density. It would have been entirely negligible compared to other forms of energy in the early Universe. Moreover, if dark energy drove inflation, what would make inflation stop?

There are, however, ways of making fluids that behave like dark energy for a short time, with a much higher effective energy density. To understand how, we need to introduce the notion of a field.

You are probably used to the idea of particles. You might have an image in mind of little billiard balls, or you might be thinking of the structure of an atom. These are useful depictions in certain regimes, but the modern understanding of particles is that they are specific configurations of fields. So, what is a field? At the simplest level, a field is an entity that has some value (of whatever kind) at every point in space and time. For example, the electric field has a magnitude and direction at every point, and if you drop a charged particle into a region with a large electric field, you'll see the field act on the particle, pushing it along in the direction of the field.

Quantum fields, the sort that give rise to particles, are somewhat more complicated—their values aren't numbers or vectors, but a different sort of mathematical objects.[5] But still, every kind of particle corresponds to a certain configuration (*excitation*) of a corresponding field. Electrons are excitations of the electron field. Photons are excitations of the photon field. When there are two electrons in a box, the field is in a doubly-excited configuration there. You get the idea.

[5] Namely, operators.

When we talk about particles, we're usually assuming that the underlying field is very nearly devoid of energy. After all, a particle represents the smallest stable packet of energy we can give to a field. But if a field happens to have a great deal of energy, single particles are no longer a good description. In many such cases, it makes more sense to assign an average value to the field, and to study how that average value evolves with time. In some ways, this average quantity behaves kind of like a particle—it has a "position," in the sense that it has a value at any given time, and it has a "velocity," in the sense that that value is changing. In this way, a field can hold potential and kinetic energy, just like a ball rolling off of a hilltop.

That potential and kinetic energy is the key to driving inflation. Suppose that early on in cosmic history, most of the energy in the Universe is stored in a field ϕ, the *inflaton* field. The average value of ϕ, which I'll write as $\langle\phi\rangle$, is thus sitting at the top of a very tall hill. That potential energy behaves just like dark energy: it has a uniform density in space, and drives exponential expansion of the Universe. But if $\langle\phi\rangle$ starts slowly rolling down the hill, converting potential energy into kinetic energy, that kinetic energy can be dissipated away. Just as a ball eventually reaches the bottom of the hill, $\langle\phi\rangle$ eventually becomes small, which brings inflation to an end.

What happens to ϕ when its average value stops rolling? The energy stored in the ϕ field needs to end up in ordinary matter and radiation, so inflation has to end with a transfer of energy into the particles we know exist later on. This process is called *reheating,* and it gets that name because it takes the *non-thermal* energy from the inflaton field and transfers into the *thermal* energy of the Big Bang plasma. The end product of the reheating process *is* the Big Bang.

With this background in mind, let's envisage the Universe during the inflationary epoch. During this time, the Universe looks incredibly sparse. Any structures or features in the landscape are being ripped apart from each other exponentially. Most importantly, most of the energy is stored in the inflaton field at this time. Right before reheating, the Universe essentially looks empty. The energy that will soon be dumped into all of the ordinary particle species hasn't been transferred yet. There may not be any matter or radiation at this time in cosmic history.

During inflation, the Universe is a formless, expanding void—a vast deep, only getting deeper, clothed in darkness. Now you can see why I draw the connection with the phrase וְחֹשֶׁךְ עַל־פְּנֵי תְהוֹם. As far as we know, there literally isn't any light in the Universe yet. In fact, speaking technically, there probably still wasn't light immediately after inflation. Some additional physical transformations needed to take place first.

6 Phase transitions: וַיְהִי אוֹר

When did light first exist? This deceptively simple question leads to some remarkable physics. To answer it, we might first consider the sources of light in the Universe today.

- The most obvious are the Sun and other stars. We'll say more about them in section 10, but they don't form until well after the Big Bang.
- There are other astrophysical sources of light, like hot gas clouds around black holes (quasars). These, too, take considerable time to form.
- There is background radiation pervading the Universe, which originated a few hundred thousand years after the Big Bang. More on that in **??**.

In each of these cases, the light itself is produced by something hot. Hot things emit light in the form of *blackbody radiation.* This is why it makes sense to say that something is "red-hot."[6] The environment of the Big Bang was tremendously hot, so it was also filled with light. (In fact, the background radiation in the Universe is just the leftover light from this time.)

So, light was present at very early times in the history of the Universe. But that doesn't answer the question of when light first came to exist. In order to answer that, we have to understand what "light" actually is.

Mathematical digression

You probably have an intuitive idea of what light is: it's what your eyes see. It's some kind of energy that propagates from place to place, and that interacts with matter, e.g. by being reflected or absorbed. You may also have a notion that light is a particle, or a wave, or both, and that it has something to do with electromagnetism. All of these statements are more or less correct. Light is a propagating excitation of the electromagnetic field. In classical electromagnetism, as it's taught to first-year university students, this means *waves* in the electric and magnetic fields, $\mathbf{E}$ and $\mathbf{B}$.

Let's get a little more advanced. The electric field $\mathbf{E}$ is a vector with three components, but it can be written in terms of just one function: the electrostatic potential φ. If φ is like a landscape with hills and valleys, the vector $\mathbf{E}$ is just the directional slope at any point on the landscape, pointing up the hill. Mathematically, $\mathbf{E} = \nabla\varphi$. The magnetic field $\mathbf{B}$ can similarly be written in terms of a *vector potential.* The vector potential $\mathbf{A}$ has a magnitude and direction at every point, and the magnetic field is a measure of how much *circulation* there is in the $\mathbf{A}$ field, i.e., how much the $\mathbf{A}$ vectors curl around any given point in space. Mathematically, $\mathbf{B} = \nabla \times \mathbf{A}$. This means that the electric and magnetic fields can be reduced to a single four-component object, the *four-vector potential,* combining φ and $\mathbf{A}$:

$$A_\mu = (\varphi,\ A_1,\ A_2,\ A_3)_\mu\,. \tag{6.1}$$

[6] "Red" is actually the color of light emitted by the coldest visible-light–emitting objects. At lower temperatures, the emitted light is infrared. As such an object gets hotter, the emitted light moves further towards blue, and eventually becomes ultraviolet.

The Greek letter μ is an index that can take the values 0, 1, 2, and 3. By convention, we write four-component vectors vectors like this one with Greek-letter indices as a reminder of their special properties.[7] I will spare you the discussion of what makes A_μ the more "fundamental" object than φ, $\mathbf{A}$, $\mathbf{E}$, or $\mathbf{B}$. Suffice it to say that when writing a theory that is compatible with relativity, A_μ is the object that matters. Light waves, these simultaneous oscillations in $\mathbf{E}$ and $\mathbf{B}$, can be reinterpreted as waves in A_μ.

In classical electrodynamics, A_μ is the basic field, the object from which all other electromagnetic phenomena derive. In quantum electrodynamics, we have a different name for A_μ: it's the *photon field*.[8] Those light waves in A_μ are the classical analogue of the quantized excitations of A_μ, or photons—particles of light.

At this point, you may expect me to say some words about how we reconcile the idea of a photon as a particle with the wavelike behavior of light. However, all of the business of particle–wave duality isn't unique to light. It applies to any quantum field, and thus to any kind of particle. We have no need to get into it here, except to say that light as we know it is made of photons, and those photons are the elementary excitations of the photon field A_μ.

Vacua and symmetries

Now, let me ask a question that seems at first to be ill-posed: when did the photon field A_μ first come to exist? It's bizarre to think that there was a time when one of our fundamental fields didn't exist, and, indeed, "exist" is not a great word for this. As far as we know, the fundamental objects that underlie the existence of light—what we call the *ultraviolet degrees of freedom*[9]—have existed throughout the history of the Universe. But what we call light is a very particular configuration of these degrees of freedom that only exists naturally at low energies, and such a configuration could not always readily arise.

To make this precise, we must introduce the notion of a *vacuum.* I have said that particles are "excitations" of a field, but for there to be a lowest-energy excited state, surely there must be a well-defined notion of a completely *un*excited state. We call this state the *vacuum* of the field. The name derives from the fact that this is the state with *zero* particles.[10] But it must be understood that the vacuum is not some canonical construction that remains unchanged for all time. The vacuum is simply the background state around which we consider excitations, which we classify using the language of particles.

How does this language work? Particles describe the excitations around the vacuum state in much the same way that x and y axes describe coordinates on a plane—i.e., as a basis for more general excitations. Here you might object that x and y axes could hardly be analogous to different species of particles, since they aren't actually different: we can simply rotate our

[7] Technically, we use this notation for *Lorentz tensors,* and the Greek letters are called *Lorentz indices.*

[8] It's also the electromagnetic gauge field, but that doesn't sound as nice.

[9] "Ultraviolet" here means high-energy or short-distance.

[10] Actually, vacua are not quite well defined in a curved spacetime, which underlies the phenomenon of gravitational particle production: different observers in a curved spacetime need not agree on the particle content of a given physical state, just as different observers in flat spacetime (i.e., in the domain of special relativity) need not agree on distances and time intervals.

coordinate system and mix the two coordinates into one another. This is a hint of a more general rule. Suppose we have some transformation under which the laws of physics must not change—in this case, spatial rotations. Such a transformation is called a *symmetry* of the system. If a given quantity is not *invariant* under the symmetry, then we should not use it to label the physical content of the state, because different observers could disagree on the value of that quantity, in the same way that two observers should be allowed to use different x and y coordinates. A point has an x coordinate according to one observer, but that doesn't mean that the value of the x coordinate is an intrinsic physical property of that point!

For the same reason, we must be careful about what excitations we select as the elementary particles of a system, or equivalently, how we choose a basis for the space of fields. We would not want to choose a set of particles that are transformed into one another by a symmetry transformation. If we did so, then different observers could fairly disagree about the number and types of particles in the system, and the labeling scheme would lose its value. Thus, we distinguish elementary particles as those groups of degrees of freedom which do *not* mix with others under symmetry transformations.

But this definition depends critically on what the symmetries actually are. I gave the example of spatial rotations, but there are many other types of symmetries among the different components of field space. Moreover, there is a simple way to *change* the symmetry structure of the elementary excitations, and this is simply to change the symmetry structure of the vacuum state. *If the vacuum state changes, so too does our notion of particles.*

Potentials and symmetry breaking

Let us now understand *how* the vacuum state can change, and what this has to do with symmetries. In order to make sense of this question, we need to first understand what defines the vacuum. I said before that the vacuum corresponds to the lowest-energy state, and if we adopt a classical picture, we can visualize this in terms of the *potential energy* that is associated with certain configurations of the field.

For a classical analogy, let's consider a ball rolling on a hilly landscape, as we did in section 5. The lowest-energy state of the ball is the state in which it is at rest at the lowest point on the landscape. That lowest point on the landscape defines the vacuum, and that's where the ball will tend to roll. Accordingly, we can picture the ball as "rolling" not in a geographic landscape, but in an abstract *energy* landscape. The ball will tend to roll towards points of lower potential energy. In fact, a classical force is nothing but the negative of the gradient of the potential energy,

$$\mathbf{F} = -\nabla V, \tag{6.2}$$

meaning that forces push objects in the direction of lowest potential energy. If we want to understand the dynamics of the ball, we need only know the landscape of potential energy. If we want to understand small perturbations of the ball around its vacuum state, we need only know the details of the potential energy around the lowest point on the landscape. The story is similar for fields, but instead of parametrizing the energy landscape in terms of the ball's

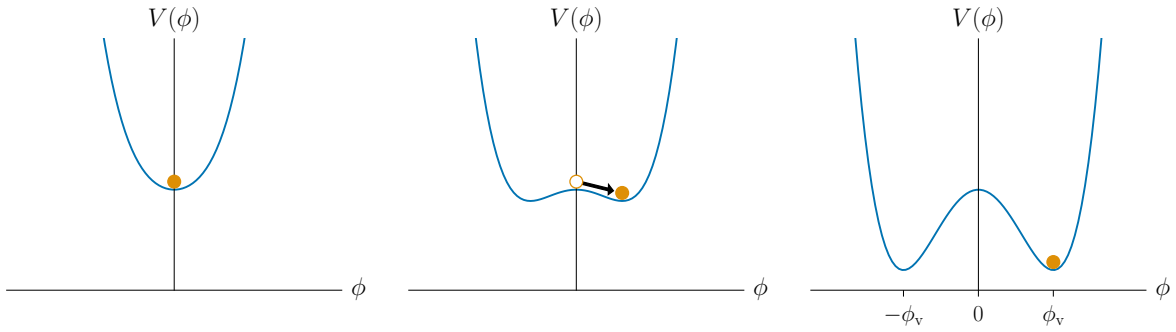


Figure 2. Spontaneous symmetry breaking. Each plot shows the potential energy as a function of the field value, ϕ, for a different temperature, decreasing (cooling) from left to right. In the first panel, the vacuum is symmetric under $\phi \leftrightarrow -\phi$. But in the second panel, a phase transition occurs, and the field chooses one of two equivalent nonsymmetric vacua. In the third panel, the field has settled into the vacuum at $\phi = \phi_v$, not into the degenerate vacuum at $\phi = -\phi_v$.

position in physical space, we know the potential energy as a function of the configuration of the field.

However, this energy landscape is subject to change. In particular, in a *thermal* system, this potential energy depends on the temperature. Modest changes in the shape of the potential can lead to large changes in *which* configuration corresponds to the vacuum state. For example, consider the potential illustrated in fig. 2. Here we are pretending to have a field whose value is just a number at each point. This cartoon shows the field value on the horizontal axis and the potential energy on the vertical axis. In the leftmost panel, at high temperatures, the vacuum state corresponds to the field having the value *zero*. This is where the potential energy is lowest. But in the rightmost panel, the potential energy can be lower for field values on either side of zero.

This figure also illustrates the change in symmetry. In the leftmost panel, the system is completely identical if we reverse the direction of the horizontal axis: that is, $\phi \leftrightarrow -\phi$ is a symmetry of the system. But in the rightmost panel, while the potential itself is symmetric under this transformation, the vacuum state is not. Here, there are actually *two* vacua with the same minimum energy. The system can only inhabit one of the two vacua, so it has to choose. No matter what choice is made, the $\phi \leftrightarrow -\phi$ symmetry is lost. This is called *spontaneous symmetry breaking.*

You are likely familiar with some simpler forms of spontaneous symmetry breaking. For example, when water freezes into ice, it forms a crystal lattice that has consistent axes, at least over short distances. Nothing about the laws of physics causes the ice to prefer any one orientation for this lattice—after all, the Universe is rotationally symmetric. But once ice forms, it has to choose an orientation, and thus the rotational symmetry must be spontaneously broken. Now we can have new phenomena that would not have been possible in the rotationally-symmetric state of liquid water. The same thing happens for particles and fields when symmetries are spontaneously broken. Moreover, we know that water turns to

ice via a *phase transition* that takes place at a certain temperature. The same can be said for different *phases* of a system of fields.

The Higgs mechanism and the first photons

The potential illustrated in fig. 2 is a toy model, meant only as a simple example to demonstrate how spontaneous symmetry breaking can take place as a system cools down. But it's actually not very different from the potential that governs the behavior of one of the fields in our Universe: the *Higgs field.* The evolution of the Higgs field, in turn, has significant consequences for the nature of light and matter in our Universe.

Let's repeat the essential feature of our toy picture of spontaneous symmetry breaking: the potential as a whole has a certain symmetry ($\phi \leftrightarrow -\phi$), but since there are multiple vacua, each individual vacuum need not exhibit the same symmetry. If the system randomly chooses a symmetry-violating vacuum, then the field can behave in a way that breaks the $\phi \leftrightarrow -\phi$ symmetry, even if that symmetry is a fundamental property of the underlying theory. This turns out to be extremely useful in understanding the behavior of particles (or fields) in our Universe. The fields in our Universe appear to respect a slightly more complicated symmetry called *electroweak gauge symmetry,*[11] except for one important feature: certain particles have nonzero masses, even though electroweak gauge symmetry says that they should be massless. Where does this apparent violation of electroweak gauge symmetry come from? It comes from spontaneous symmetry breaking! At low temperatures, the potential of the Higgs field exhibits a set of equivalent symmetry-breaking vacua, and the system chooses one, just as in our toy example. This background value of the Higgs field is a symmetry-breaking quantity that can generate these otherwise forbidden masses. This is called the *Higgs mechanism.*

The effects of electroweak symmetry breaking via the Higgs mechanism have consequences that go beyond giving masses to massless particles. Since this changes the vacuum state and symmetries of the system, it also *changes what the elementary particles are* for the set of fields that change under electroweak gauge symmetry transformations. As the prefix "electro-" might hint to you, this includes photons. Photons are only a well-defined excitation of the system *after* electroweak symmetry breaking. Prior to electroweak symmetry breaking, there isn't an elementary excitation of the system that corresponds to what we think of as light. Instead, the elementary excitations combine electromagnetism and the fields governing the weak interactions—hence "electroweak." At high temperatures, these forces are unified: you can't tell the difference between a photon and an excitation corresponding to the weak interaction, because the two are related by electroweak gauge symmetry. To say that they are different would be akin to claiming that a photon moving North is a different species of particle from a photon moving West. This is clearly false because a simple rotation—a symmetry of the Universe—transforms one into the other. Similarly, it makes no sense to

[11] For the mathematically inclined, every field is associated with a representation of SU(2), which defines the symmetry transformations. That is, configurations that are related under this group action are considered to be physically identical.

speak of photons as being distinct entities. Instead, the natural elementary excitations of the system are *combinations* of photons and other particles.

This is all an extraordinarily long-winded way of saying that in the very early Universe, at extremely high temperatures, there is no such thing as "light," in the way that we experience it. A separate electromagnetic field, photons, and all of those familiar constituents of the Universe only exist after the temperature drops and electroweak symmetry breaking takes place via the Higgs mechanism, about 10^{-12} seconds after the Big Bang. Once this happens, for the first time, there is light.

7 Hadronization and nucleosynthesis: וְרֹאשׁ עַפְרוֹת תֵּבֵל

The breaking of electroweak symmetry in the early Universe is not the only phase transition that takes place in the first second after the Big Bang. Another such transition gives rise to the basic constituents of most matter, protons and neutrons, which don't exist immediately after the Big Bang.

Before explaining the change that took place in the early Universe, let's review the basic structure of matter in the Universe today. For the moment, I will completely ignore the dark matter that I mentioned in section 3, and talk only about the ordinary matter that makes up visible objects—including you. In fact, let's use you as our example, and zoom in progressively. Zooming deep into a single cell, the underlying structural units are molecules. Molecules come in many different forms, corresponding to different combinations and arrangements of atoms. For present purposes, let's suppose we zoom in on a molecule of glyceryl trioleate, one of the common fat molecules. This molecule is composed of 167 atoms bound together ($C_{57}H_{104}O_6$). The chemical bonds that hold the molecule together are governed by nothing but electromagnetism, which behaves a little differently here in the quantum mechanical regime. Let's zoom in further onto one of the Carbon atoms. This atom consists of six negatively-charged electrons surrounding a positively-charged *nucleus*, or core. Let us now ask how it got to be that way.

Hadronization and the confinement phase transition

The word "atom" derives from the Greek word for "indivisible," which is unfortunate nomenclature, because the atom is anything but. We can quite easily *ionize* the atom, removing the electrons and leaving only the positively-charged nucleus. Nor is that the end of the chain of divisions. The nucleus is composed of protons and neutrons. Each proton is positively charged, exactly opposite an electron, so there are six protons in the Carbon nucleus. There are also six neutrons.[12] This is *still* not the end of the story. The protons and neutrons also have internal structure. They are assembled out of different types of *quarks,* which, for various historical reasons, have quirky names: up, down, charm, strange, top, and bottom. A proton is made from two up quarks and one down quark, while a neutron is made from one up quark and two down quarks.

As far we know, quarks are elementary—any underlying structure therein is too small for us to see. Regrettably, however, that they are elementary does not mean that they are simple. Making predictions about quarks and their combinations is one of the greatest challenges in physics today. This is because unlike other particles, quarks are subject to the *strong interaction.* Almost everything that we compute in particle physics is computed *perturbatively,* i.e., by assuming that the interactions make only small corrections to the noninteracting behavior of the particles. With quarks, the interaction is too strong, and this picture completely breaks down. This means that there are still some phenomena in the

[12] The number isn't always the same, even within a given element. Here we are looking at ^{12}C, but ^{13}C has seven neutrons, and the unstable isotope ^{14}C has eight.

behavior of quarks that we have yet to fully understand. But there are many things that we are quite sure of.

One of the most remarkable behaviors of quarks is a phenomenon called *confinement.* In short, confinement is the property that individual quarks cannot exist stably on their own. The energy that binds two quarks together is so great that it exceeds the mass of a quark, which means that it is energetically favorable for a free quark to produce a partner out of the vacuum rather than to exist alone. So, they only exist stably in *hadrons,* or combinations of quarks, such as protons and neutrons. We only ever see individual quarks on extremely short timescales before they *hadronize* and form a new combination of quarks.

Thus, for all intents and purposes, when describing matter in the Universe today, we may as well think of protons and neutrons as the elementary units of matter—we will never find a quark floating around in space. However, for processes at extremely high energies, or on extremely short timescales, the interactions of the quarks diminish rapidly, and they behave just like free particles again. This is called *asymptotic freedom,* and it might make you wonder: what happens when hadrons are heated up to a very high temperature, as they were in the early Universe?

In fact, when a medium of hadrons is heated up to a temperature above the scale of the strong interaction (about 100 MeV, or roughly one trillion degrees), the quarks *do* behave as though they are stable free particles. The hadrons dissolve, and the quarks form a new state of matter called a *quark–gluon plasma.* This was the Universe after the Big Bang, even after electroweak symmetry breaking: a primordial hot soup of quarks and gluons. It was only after the first microseconds had passed that the soup cooled enough to congeal into hadrons, in what we call the *confinement phase transition.* Only at this point did the Universe contain the microscopic constituents of what would become matter as we know it.

The phrase in the title of this section is from Proverbs 8:26 / משלי ח׳:כ״ו. With some surrounding context, it reads:

> I, wisdom, dwell with prudence, and find knowledge and discretion…I was set up from everlasting, from the beginning, before ever the earth was…while as yet he had not made the earth, or the fields, or the first of the dust of the world.

אֲנִי־חָכְמָה שָׁכַנְתִּי עָרְמָה וְדַעַת מְזִמּוֹת אֶמְצָא:
…מֵעוֹלָם נִסַּכְתִּי מֵרֹאשׁ מִקַּדְמֵי־אָרֶץ: …עַד־לֹא
עָשָׂה אֶרֶץ וְחוּצוֹת וְרֹאשׁ עַפְרוֹת תֵּבֵל:

If you were looking for what *the first of the dust of the world* could mean, I suggest that you could do worse than the first hadrons.

Big Bang nucleosynthesis

To form hadrons is not the same thing as form nuclei. At this point in cosmic time, we now have protons and neutrons. The protons are technically Hydrogen (^{1}H) nuclei, so we have only the very simplest isotope of the very simplest element. That would be problematic for the Universe as we see it today: even though most elements heavier than Hydrogen are

produced later by stars (section 10), the stars themselves wouldn't form at the same rate in the absence of helium.

Fortunately, the lighter elements can form directly in the hot Big Bang plasma, essentially by fusion. The simplest nucleus after ^{1}H is Deuterium (^{2}H), which combines one proton with one neutron. After forming Deuterium, one can add a proton to form ^{3}He, the lightest stable isotope of Helium, or one can add a neutron to form Tritium (^{3}H). But nothing interesting can happen until Deuterium forms, and this can't happen until the temperature of the Universe drops below the binding energy of a Deuterium nucleus, which is about 2 MeV. The Universe reaches this temperature around 1–10 seconds after the Big Bang. As the temperature drops, the hot plasma produces quite a bit of Helium. We can make precise computations of the abundances of different nuclei at this point, and we find that about 25% of the matter (by mass) ends up in the form of ^{4}He, with the rest left as Hydrogen. (This matches observations exquisitely well, by the way.) A very small amount of Lithium is produced as well, and an even smaller fraction of the matter ends up in Beryllium.

This process of *Big Bang nucleosynthesis* sets the stage for the drama of stellar chemistry that follows hundreds of millions of years later. However, what we have at this point are still not quite atoms as we mostly encounter them today: they are completely ionized, and they remain so until the next stage of cosmic evolution.

8 Decoupling and recombination: וַיַּבְדֵּל אֱלֹקִים בֵּין הָאוֹר וּבֵין הַחֹשֶׁךְ

The next step in cosmic evolution that I want to highlight here is called *recombination.* This is when electrons "recombine" with the newly formed nuclei to form the first neutral atoms.

First of all, I should point out that *re*combination is a misnomer: the electrons were not combined with the nuclei at any previous time in cosmic history. But I want to say a bit more about the physics of recombination, since thinking about it only as "ions and electrons forming atoms" doesn't do justice to its cosmological significance.

Why did recombination take place at a particular time? As with all the previous steps in cosmic evolution, it's a matter of temperature: the Universe had to cool to a temperature that was low enough for neutral atoms to survive without being immediately ionized again. But this time, there is also a more tangible way to understand the process. Recall that the plasma of the early Universe was *thermal.* This means, in particular, that it glowed, just like any other hot object. When the temperature of the Universe was, say, 100 eV, this means that the plasma was constantly emitting and absorbing photons with energies of roughly 100 eV. Such a photon will readily ionize atoms. A Hydrogen atom, for example, can be ionized with just 13.6 eV of energy. So, it should come as no surprise that the temperature of the Universe had to drop below that mark before neutral Hydrogen could form. The detailed considerations of thermal equilibrium imply that the Universe actually had to cool to temperature of around 0.3 eV before most of the Hydrogen combined with electrons and became neutral.

Remarkably, during this epoch, you would have been able to see the glowing medium with your own eyes while you disintegrated. Photons at 0.3 eV are infrared, meaning that they are not energetic enough for humans to see them. But since the glowing Universe would behave like a blackbody, there would be considerable variance in the energies of the photons. Visible light corresponds to photons with energies between about 1.5 and 3 eV, i.e., between 5 and 10 times the temperature of the Universe at recombination. From the blackbody radiation law, one can readily compute that about 25% of the energy in the radiation bath is actually made up of photons with energies between 5 and 10 times the temperature—visible light! At this point in cosmic time, on the order of 1% of all the energy in the Universe is in the form of visible light. By my rough estimate, the visible light in any direction would look about a thousand times brighter than the Sun.

The fact that atoms became neutral is important for several reasons, but the most striking of them is that for the first time, the Universe became transparent. Electrically-charged particles scatter light, so before recombination, light couldn't travel very far without encountering an electron or a charged ion and bouncing off. Once recombination took place, and all of the charge settled into net-neutral atoms, light could travel unimpeded.

If you were floating in the Universe at the epoch of recombination, wearing a very sturdy spacesuit, you would see a dense fog of bright light, not unlike what you see when you switch on your high beams on a foggy evening—the light gets scattered in the same way. But as recombination took place, the fog would recede, and you would be able to see some distance in front of you. Now, here's where the finite speed of light makes this interesting: if you looked

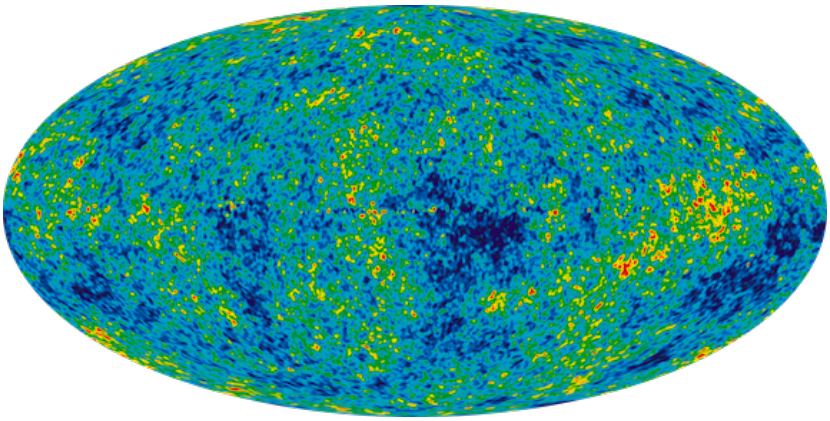


Figure 3. The cosmic microwave background as measured by the WMAP satellite [2]. Apart from false color, this image also subtracts away the doppler effect from the motion of our galaxy.

far, far away on the horizon, say, a light-year away, you would need to wait a year before that spot would appear to go through recombination and become transparent.[13] Apply this same reasoning everywhere, and it implies that the fog would appear to recede away from you at the speed of light. So, even today, if you could stare very far away, and see light from billions of light years away, you should see the edge of that same fog, which has by now been receding for about 13.7 billion years.

We can do this experiment, and we can actually see this dim fog, shown in fig. 3. The light as we see it now is called the *cosmic microwave background* (CMB).

The CMB is sometimes said to be a "baby picture" of our Universe. That's pretty accurate. This light is coming to us directly from the epoch of recombination, which took place about 300,000 years after the Big Bang. We can't see any further back in time—the fog obscures it. But like any good baby picture, the CMB gives us an enormous amount of information about the conditions of the early Universe.

I chose to start this section off with a reference to the creation story: וַיַּבְדֵּל אֱלֹקִים בֵּין הָאוֹר וּבֵין הַחֹשֶׁךְ, "And God separated between the light and the darkness." Before recombination, light and matter were in some way inseparable due to the frequent scattering between photons and charged particles. In particular, this meant that the two maintained a weaker form of the same thermal equilibrium that characterized the early Big Bang plasma. It was only at this moment that the two fluids, one of light and one of matter, *decoupled* from one another, allowing them to evolve separately.

Dark matter decoupling

At this point, I would be remiss if I failed to mention a similar hypothetical process—a different kind of decoupling—that would also surely merit the label of separation between

[13] We are pretending for the purposes of this visual that recombination happens instantaneously. It does not, when compared with a year, but it does in comparison with the rest of cosmic time.

light and dark, if it took place. This is the decoupling of dark matter from ordinary matter.

I will skip almost all of the remarkable physical context for this process, and jump straight to the punchline: in many of the most popular models for dark matter—which is still not well understood—the dark matter species was part of this early thermal equilibrium. If the interactions between ordinary matter particles and dark matter particles are just strong enough, this would happen naturally: ordinary matter particles would annihilate into dark matter particles, and vice versa. As the Universe cooled, this process would become rarer and rarer and would eventually stop altogether. This would leave behind a separate dark matter plasma, evolving on its own, decoupled from ordinary matter.

If this did happen, it would have happened at much higher temperatures, probably close to the electroweak phase transition. I want to stress that we don't know whether this happened. It's a natural consequence of some models, and has some interesting explanatory power for the abundance of dark matter in our Universe today. The answer will have to wait. But I also want to stress that regardless of whether dark matter itself was part of the equilibrium bath of the early Universe, these *decoupling* events are crucial to the structure of the Universe as it is today. The Universe today is not at all in thermal equilibrium! It has gotten to be this way by a sequence of decoupling events that each imprinted interesting information about cosmic history onto the late Universe.

One of the tantalizing features of decoupled species is that they transmit information to us from the time of their decoupling. We can't see light from before recombination, but the light that we receive from recombination tells us about the conditions of the Universe at that time. If we were to discover that dark matter indeed decoupled from the plasma at an earlier epoch, then the dark matter in the Universe today would contain information about the Universe at that primordial time. The same statement applies to any species that was not in thermal equilibrium. Apart from dark matter, the most important example is furnished by gravitational waves, i.e., ripples in spacetime. These waves were never in equilibrium because of how weakly they interact with matter. This means that certain types of gravitational wave signals today might tell us about the Universe at extremely early times, revealing, for example, the physics of inflation. Decoupling separates light from dark, but we may yet be illuminated by the dark.

9 Structure formation: יְהִי רָקִיעַ בְּתוֹךְ הַמָּיִם

One of the most important lessons we learn from the CMB is that the Universe at 300,000 was incredibly homogeneous. Hotter and denser regions of the Universe were hotter and denser than the background by only a factor of 1.00001, on average. This is, of course, in line with predictions for an equilibrium thermal soup.

But the Universe today looks nothing like that. Matter is clustered in massive galaxies separated by enormous voids. Rather than homogeneity, our Universe exhibits an abundance of *structure.* What happened between then and now?

The very simple answer is "gravitational collapse." The slightly denser regions in the early Universe would gravitationally attract slightly more matter, and thus become denser, and thus attract more matter, and so on. But this picture still leaves a few important questions. Why doesn't the pressure of the hot gas resist the infall of more matter? What is the end state of gravitational collapse? And, even more importantly, how does this collapse process compete with cosmic expansion?

To answer these questions, we must return to the idea of *dark matter,* which I first mentioned in section 3. Recall that dark matter accounts for about 80% of all matter in the Universe. This is a good moment to explain why dark matter is "dark." Actually, it should be called "invisible." Of all of the interactions that we observe in ordinary matter, dark matter participates only in the gravitational interactions. This doesn't mean that there aren't other interactions that we haven't yet discovered, but it does mean that the sum total of the interactions between dark matter particles and other particles—ordinary or dark—are extremely weak. To a good approximation, at least, dark matter behaves as though it were a fluid with only gravitational interactions.

This is extremely important for the behavior of gravitational collapse. Amongst ordinary particles, repulsive electromagnetic interactions slow down the collapse of gas. In fact, even today, when mapping galaxies, we can see that there is a certain distance scale between pairs of galaxies that occurs much more frequently than random chance would predict. This distance corresponds to the wavelength of pressure waves—literally *sound waves*—in the matter fluid of the early Universe.[14] We see galaxies separated at this characteristic distance for the same reason that ripples on a lake are separated by a certain distance scale. So, if these interactions are powerful enough to impart these sound waves into the distribution of galaxies, how did galaxies form at all without being destroyed by the pressure? As it turns out, collapse of the gas is crucially enabled by collapse of the dark matter.

The gas cannot begin to collapse until recombination. Until that point, the tight coupling between the matter and the radiation in the Universe—the "dense fog" of section 8—means that radiation pressure can push gas out of overdensities, and force it into underdensities. But the dark matter does not exhibit the same interactions, and is *decoupled* from radiation well before this epoch. For the first 300,000 years of cosmic history, while the gas stayed more or less homogeneous, the dark matter was collapsing into dense clouds that would become

[14] These sound waves are called *baryon acoustic oscillations.*

the seeds for cosmic structure. As soon as the gas decoupled from radiation and could finally collapse, it was pulled in by these dark matter clouds.

By the way, this turns out to be essential to forming galaxies *at all* in our Universe. Remember, the Universe today is just starting to expand exponentially, which further impedes the collapse of gas. Without dark matter, galaxies wouldn't form before this exponential expansion would become dominant, ripping apart any nascent overdensities. *We are here because of dark matter collapse.*

Now, what happens after dark matter has collapsed for a long time? Ordinary matter makes objects like stars and planets. Does dark matter do that? The answer is, probably not—or, at least, most of the dark matter does not. We will see in section 10 that these familiar objects only form thanks again to the nongravitational interactions of matter particles, which dark matter lacks. What happens instead? You might think that there would be no end to the collapse. As the dark matter gets denser and denser, surely it would pull the rest of the dark matter closer and closer together, making it more and more dense, eventually forming a black hole. Most of the dark matter doesn't do this, however. The reason is simply conservation of energy and angular momentum.[15] Even if the center of the cloud becomes extremely dense, the typical energy and angular momentum of the dark matter particles means that they *orbit* the center rather than falling in.

At a certain point, when the kinetic energy and potential energy are balanced, the collapse process stops, and we are left with a stable dark matter cloud.[16] These clouds, or *halos,* are the settings for galaxy formation. They merge together over time to make the nested structures we see in the Universe around us: smaller halos with smaller galaxies are arranged within larger halos containing larger central galaxies. This is why, for example, our own Milky Way galaxy has several smaller "satellite" galaxies. The description here sounds qualitative, but as with most of these things, it can be made quantitative. The statistical distribution of galaxies and their relationship to dark matter halos can be made exquisitely precise, and it is one of our best diagnostics of the properties of dark matter today.

I chose to link this section to the following line:

וַיֹּאמֶר אֱלֹקִים יְהִי רָקִיעַ בְּתוֹךְ הַמָּיִם וִיהִי מַבְדִּיל בֵּין מַיִם לָמָיִם׃

And God said, There shall be an extended structure spread out in the midst of the water, and it shall divide between water and water.

The word רקיע is often translated as "firmament." I attempted a more creative translation above. The word connotes something that is spread out and thinned, and is used elsewhere to describe the manufacture of metal plates and foils. Such a spreading and thinning is actually the first step of dark matter collapse: overdensities first collapse into sheets, and then they form filamentary and finally cloudlike structures separated by vast voids.

[15] As I've mentioned a few times now, these conservation laws don't work on cosmological scales, but they are very effective approximations on the scale of a collapsing dark matter cloud.

[16] This is called *virialization,* since the balance between kinetic and potential energy is dictated by the virial theorem of classical mechanics.

10 Stars, galaxies, and quasars: יְהִי מְאֹרֹת בִּרְקִיעַ

At this point, we have described the evolution of the Universe from the earliest moments through recombination, and we have described how structures start to form thereafter. It is now time to ask how sources of light arise in the Universe.

Sometimes people object to the literal reading of בראשית because light comes into existence before the Sun. This is a ludicrous reason to take issue with the Biblical chronology, especially considering that there are so many better bones to pick. The reason why this is so silly is that there is in fact a great deal of light in the Universe before stars form, originating directly from the hot Big Bang plasma. That said, this primordial light quickly fades as the Universe expands, and it is extremely faint today. So now, let's ask: what was the next source of light in the Universe?

Actually, considerable time passed between recombination and the formation of new light sources. For this reason, the epoch between recombination and the first emissive objects is called the "dark ages." Structure is slowly and silently accumulating during this period, but it takes several hundred million years for the first stars to form. In order for stars to form, the primordial gas needs to *cool.* You might think that stars should form from hot gas, but the bigger challenge at this stage is to get the gas to collapse into a small volume. Hot gas resists collapse. So, it takes hundreds of millions of years of structure formation and gas cooling processes before the gas clouds fragment and collapse into stars.

These first stars, called *Population III* stars, are very different from the stars we see today. They form directly from the primordial gas, which is extremely different from the intergalactic medium today. The gas left over from the Big Bang plasma has a chemical composition determined only by Big Bang nucleosynthesis (section 7), meaning that it is almost entirely a mixture of Hydrogen and Helium. This results in massive stars, ∼1000 times more massive than the Sun, that burn brightly for a short time—just a few million years. As such, everything we know about Population III stars is inferred from our knowledge of cosmology. We have never seen one directly.

Although we have never seen Population III stars, they play an extremely important role in the evolution of the Universe: they kickstart the process of *chemical evolution.* The primordial gas is all Hydrogen and Helium, but the Universe today contains many heavier elements. These elements come from stellar evolution: stars fuse together light elements to form heavier ones. The first stage of this process takes place in Population III stars. Their explosive deaths seed the cosmos with the first *metals*—another misnomer. In astronomy, a "metal" is any element heavier than Helium. Their significance lies in their role in catalyzing star formation.

Stars only form when gas can cool. Cooling relies on the ability of gas to radiate away excess energy via the electromagnetic interaction. (Remember, this is why dark matter doesn't form stars.) But metals are far more efficient at radiating away heat than Hydrogen, so as the abundance of metals increases, so does the star formation rate. Stars process the lighter elements into metals, increasing the metallicity of the gas, and creating the conditions for

another generation of stars—*Population II.* Population II stars have low levels of metals, but this is enough to radically alter their evolution. These stars can be long-lived, and many of them are still with us today. They likewise increase the metallicity of their environments, which gives rise to high-metallicity Population I stars like our Sun.

Perhaps it is obvious, but it is worth special mention: by mass, Earth and our own bodies are composed almost entirely of "metals" heavier than Helium. This means that our own constituent elements were largely produced in stars, and have come to be here only by virtue of the deaths of those stars. We literally are stardust.

As stars form in galaxies, we also see the emergence of another emitter of light: *quasars.* Quasars are massive black holes. You might not be used to thinking of black holes as light sources, but they can actually among the brightest objects in the Universe. Black holes grow by *accretion,* i.e., by swallowing gas. But in order for the gas to fall into the black hole, it must again lose energy and tighten its orbit. This means that as black holes accrete gas, the gas radiates away all of its orbital energy. This turns out to be an extremely efficient mechanism to produce radiation, as these things go, and quasars are extremely bright as a result. The brightest quasars are thousands of times brighter than our entire galaxy.

The quasars are so bright, in fact, that they change the properties of the gas everywhere in the Universe. Recall that between Big Bang nucleosynthesis and recombination, all of the atoms in the Universe were ionized, so the medium behaved like a plasma rather than like a neutral gas. But now that quasars (and early stars) begin lighting up the Universe, producing enormous numbers of high-energy photons, the gas becomes ionized again. This process is called *reionization.* By the time one billion years has elapsed since the Big Bang, the intergalactic medium is almost completely ionized once again throughout the Universe. The ionization of the gas in turn influences subsequent the evolution of stars and galaxies, which now becomes bottlenecked by feedback. This is critical to understanding the population of galaxies we see in the Universe today.

The quote in the title of this section needs no explanation: יְהִי מְאֹרֹת בִּרְקִיעַ, or, to use my translation from the previous section, "There shall be light sources in the extended structure."

11 Planet formation: יִקָּווּ הַמַּיִם מִתַּחַת הַשָּׁמַיִם אֶל־מָקוֹם אֶחָד

It is now time to talk about the formation of the Earth itself. Unfortunately, this section will be somewhat less detailed than the others, since I'm not an expert in planetary formation.

In order to say anything about the formation of the Earth, we need to back up and address the formation of the rest of the Solar System, starting with the Sun, about 5 billion years ago. The Sun, as with other stars, could only form as gas cooled. The best place for gas to cool is in a dense environment with high metallicity, low ionization, and high abundances of molecular Hydrogen (H_2). These particles can radiate away heat much more efficiently than ionized hydrogen atoms, so they serve as coolants. In the context of the late-Universe Milky Way, these requirements are best served by giant molecular clouds. These clouds have high densities and high H_2 abundances, which makes them the ideal stellar nurseries.

The Sun indeed formed from such a cloud. As the cloud cooled, different regions within the cloud underwent gravitational collapse, i.e., it *fragmented.* One of these fragments eventually became the Solar System. As it collapsed, it first flattened into a disk. This is a general feature of collisional systems as they collapse: there is a preferred axis determined by the total angular momentum of the system (i.e., its net spin). Collisions can flatten the disk in that direction, but the disk can't readily shrink inwards because of angular momentum conservation. This was the *protoplanetary disk* of the Solar System, from which the Sun and planets formed. The disk-like shape is also the reason for the nearly planar shape of the Solar System today.

The center of this disk eventually became hot and dense enough under gravitational collapse to ignite a star—our Sun. The small dust grains collected slowly, eventually forming *planetesimals* about the size of a city. These planetesimals collided and merged to eventually form planets, including Earth. The Moon, however, did not form by this process. Instead, it is hypothesized that the Moon was produced as the result of a collision between Earth and another planetary body later on in the history of the Solar System.

Immediately after formation, the Earth was extremely hot due to the dissipation of energy in collisions during the accretion process. The surface of the Earth was fully molten, and did not solidify into the crust as we know it until it cooled. (Actually, the giant impact that formed the Moon likely remelted a large portion of the Earth's surface.) You would think that liquid water could not exist in such a hot environment, but since atmospheric pressure was so high, the oceans formed quite soon, within the first few hundred million years of Earth's history.

In contrast to star formation, which relies crucially on the chemical environment and cooling and heating processes, the formation of planets is really just the accrual of matter by successive collisions, eventually assisted by gravity. This slow gathering of matter is reminiscent of the phrase יִקָּווּ הַמַּיִם מִתַּחַת הַשָּׁמַיִם אֶל־מָקוֹם אֶחָד — "The water beneath the sky shall gather to a single place." More properly, it gathers into several distinct places of several different types. However, thanks to its position with respect to the Sun, Earth is the only body in the Solar System today with a significant amount of liquid water.

12 Anthropics and thermodynamics: לֹא תֹהוּ בְרָאָהּ לָשֶׁבֶת יְצָרָהּ

The physical understanding of cosmology has resolved a huge number of mysteries, and gives us a clear picture of how the Universe came to be the way it is today. However, I want to close this discussion of physical cosmology by mentioning a couple of philosophical issues in the history of the Universe that have become *more* troubling as our physical understanding has deepened.

The first is the question of whether the Universe is finely tuned for human existence. This discussion has taken many forms over a very long time—and it is true that if some apparently fundamental constants took different values, the Universe would be inhospitable to human life. If certain parameters of the fundamental interactions were modified, chemistry would be altered to such a degree that the formation of life would be implausible. Other modifications could make it impossible to form stars, or impossible to form stable orbits, or even impossible for the Universe to exist for very long at all before collapsing or expanding exponentially and ripping all matter apart.

A common religious approach to this problem is teleological, i.e., the solution is to assume that the Universe has been structured with certain parameters for the *purpose* of supporting life. In this picture, some cause beyond the material Universe tunes the Universe with life as an end goal. This is the reasoning behind Isaiah 45:18 / ישעיהו מ״ה:י״ח, which reads:

כִּי כֹה אָמַר־ה׳ בּוֹרֵא הַשָּׁמַיִם הוּא הָאֱלֹקִים יֹצֵר הָאָרֶץ וְעֹשֶׂהּ הוּא כֹוֹנְנָהּ לֹא־תֹהוּ בְרָאָהּ לָשֶׁבֶת יְצָרָהּ אֲנִי ה׳ וְאֵין עוֹד׃

For thus says the Lord that created the heavens; God himself that formed the earth and made it; he has established it, he did not create it a waste land, he formed it to be inhabited: I am the Lord; and there is none else.

However, this is not the only solution. There is an entirely different solution that turns this reasoning on its head: perhaps the Universe must look hospitable to intelligent life *because there is intelligent life in it.* In other words, if the Universe could not accommodate intelligent life, then nobody would be asking this question in the first place! This is a crude statement of the *anthropic principle:* that our sample of one Universe may be an incredibly biased sample, because in order for anyone to see it, *someone* must first come to exist.

This is at first a trivial statement. What makes it truly interesting is the following possibility: perhaps there are other universes in which these parameters are different, maybe even randomly selected. If there are many universes constantly being born, then it would be no surprise that most of them would not contain life. But some minuscule fraction *would* contain life, by sheer chance—and in these universes, intelligent beings would one day ask why their rare Universe was apparently finely tuned for their existence.

Remarkably, the processes of physical cosmology provide the mechanism for exactly this scenario to take place. Could there be many universes? Absolutely, in the following sense: our *observable Universe* is bounded by a causal horizon, past which no light or information can move. As such, a causally-independent patch of spacetime beyond the horizon effectively constitutes a separate Universe. No information can ever be exchanged between these two patches, nor can any physical events in one patch affect anything in the other patch. Now,

here's the key idea: if our observable Universe underwent inflation, as we think it did (section 5), then inflation must have ended at some point *within our Universe.* But it did *not* need to end elsewhere. If there was some small part of the larger spacetime where the end of inflation came a little later, that patch of spacetime would have swelled to an enormous volume, and "most" of the spacetime would still be inflating. Iterating this idea, it is a real possibility that inflation never really ends, and that only small patches of the full spacetime stop inflating and make universes that look like ours.

So, perhaps there are many causally disconnected universes forming all the time. But to give this idea legs, we also need fundamental parameters to be randomly selected. This is also possible by the same mechanisms that we have talked about in these notes. In particular, it is possible that some parameter values are randomly selected by phase transitions, not unlike the electroweak symmetry breaking transition that we discussed in section 6. If "fundamental" constants are actually fixed by some unknown phase transitions, then they could take completely different values in different patches of spacetime, as different random vacua are selected in different patches. This idea becomes even more general in the context of string theory, where even the number of spatial dimensions can vary between different universes.

It therefore seems that physical cosmology provides plausible mechanisms for the existence of a large ensemble of independent universes with random values of fundamental constants. If this actually happens, then the anthropic principle is inevitable: most of those universes will never form intelligent life, but some of them will, and only in those universes will life-forms ever ask the question of how they got to be there.

There is one more deep sense in which the Universe is finely tuned—not for life, but for time to pass at all. The second law of thermodynamics is a deep principle that is really more a law of statistics than of physics. It tells us that entropy, or the number of possible arrangements corresponding to a physical state, must increase with time. Entropy is sometimes said to be a measure of "disorder" in a system. More practically, this actually serves as a definition of the direction of time, often called the *arrow of time.* When you watch a video of an egg breaking, you know when it's being played backwards: statistics clearly disfavors the reassembly of a broken egg. On the other hand, if you were to watch a much simpler system—say, a planet orbiting a star—you probably would not be able to tell whether the video was being played forward or in reverse. Such a simple system does not face this statistical penalty, and the evolution is time-reversible.

But our Universe *does* have an arrow of time, which means that entropy is continually increasing. And if entropy is increasing, this means that the entropy of the early Universe must have been incredibly *low.* How and why did this low-entropy state come to be in the first place? This is a hard question, and for the moment, we can only speculate. But to the extent that entropy corresponds with "chaos," it invites a deep question about the long future of our Universe. What made our Universe start out in such a special place, and what will it look like long after we're gone?

Part II
Metaphysical cosmology

In the previous part, I gave a qualitative overview of the modern understanding of cosmology. Now, in this second part, I want to do something very different: I want to sketch a few ways that one might draw some new meaning from the textual creation narrative in בראשית by filtering it through the perspective of physical cosmology. In particular, in the next five sections, I want to suggest five different interpretive strategies in which understanding the scientific perspective becomes a fruitful exercise.

In contrast to the previous part, which is my summary of scientific consensus, this part contains a great deal of my own opinions. However, we are far from the first people to ask these questions, and I have done my best to include a representative set of scholarly opinions from throughout history. On that note, I must warn the reader and apologize in advance: I am not a trained philosopher, and there are surely many deficiencies in the following sections.

13 The literal reading (פשט) and its metaphoric extensions

Let's begin with the simplest framework for reading בראשית scientifically: a *semi*-literal intepretation. Numerous commentators have pointed out that insisting on a purely literal reading is counterproductive. For example, in *Guide for the Perplexed,* Maimonides writes:

> אחת מהם – זאת ההקדמה, והיא שכל מה שנזכר ב׳מעשה בראשית׳ ב׳תורה׳ אינו כולו על פשוטו כפי מה שידמו ממנו ההמון [...] כי הפשוטים ההם מביאים אם להפסד דמיון גדול והרכבת דעות רעות בחוק האלוה או לביטול גמור וכפירה ביסודי התורה. והנכון – לעזוב ולהרחיק בחינתם בדמיון לבד והנטיה מן החכמות, ולא כמו שיעשו ה׳דרשנין׳

> First, the account given in Scripture of the Creation is not, as is generally believed, intended to be in all its parts literal. [...] The literal meaning of the words might lead us to conceive corrupt ideas and to form false opinions about God, or even entirely to abandon and reject the principles of our Faith. It is therefore right to abstain and refrain from examining this subject superficially and unscientifically.

Indeed, I am not aware of any major commentator prior to the 20th century who advocated for strict literalism in the interpretation of תורה in general, nor specifically in the interpretation of בראשית. A strict young-Earth creationist viewpoint has never had much purchase in Jewish thought. However, the simplest way to reconcile the text with the scientific picture is to read the text as *metaphorically* descriptive of real events in cosmic history. We will shortly attempt to do just that. First, though, even if only for sociological reasons, it is interesting to briefly consider the history of literalism in the interpretation of בראשית.

The character of theological literalism

Although literalism is not common in Jewish theology, let us ask how it has been expressed when it has occasionally surfaced. By "literalism," I mean an explicit rejection of metaphorical

interpretation. Again, this is extremely rare among the early commentators. Even Rashi, one of the more literally-minded interpreters, leaves room for the presence of deeper meanings that deviate from the literal textual meaning. My suspicion is that Jewish theologians started to show hostility towards nonliteral readings only once *competing* creation narratives emerged and purported to have empirical or philosophical support. These alternate narratives, if taken seriously, would be in tension with the literal interpretation. As such, there were two theological options: either one could reject the literal interpretation, or reject the credibility of the alternate narratives.

Most commentators took the first approach, as with Maimonides above. Indeed, in the same section of *Guide for the Perplexed* (c. 1190 CE), Maimonides even directly allows for scientific errors on the part of his predecessors:

ולא תבקש ממני שיסכים כל מה שזכרוהו מענין התבונה למה שהענין נמצא – כי החכמות הלימודיות היו בזמנים ההם חסרות, ולא דברו בהם על דרך קבלה מן הנביאים, אבל מאשר הם חכמי הדורות ההם בענינים ההם, או מאשר שמעום מחכמי הדורות ההם.

You must, however, not expect that everything our Sages say respecting astronomical matters should agree with observation, for mathematics were not fully developed in those days: and their statements were not based on the authority of the Prophets, but on the knowledge which they either themselves possessed or derived from contemporary men of science.

But some did take the opposite position. In particular, Judah Halevi's *Kuzari* (1140 CE) directly confronts Aristotle's model of an eternal cosmology. Here, there was no creation event, in manifest conflict with בראשית. He tells the story of a powerful ruler referred to as Al Khazari,[17] who invites different religious authorities and also a "philosopher" (i.e., a scientist) to argue for their various systems of cosmology. The philosopher goes first, saying:

וְאִם יֹאמְרוּ הַפִּילוֹסוֹפִים שֶׁהוּא בָרָאֲךָ הֵם אוֹמְרִים זֶה עַל דֶּרֶךְ הַעֲבָרָה, מִפְּנֵי שֶׁהוּא עִלַּת הָעִלּוֹת בִּבְרִיאַת כָּל נִבְרָא, לֹא מִפְּנֵי שֶׁהוּא בְּכַוָּנָה מֵאִתּוֹ. וְלֹא בָרָא מֵעוֹלָם אָדָם

If philosophers say that He created thee, they only use a metaphor, because He is the Cause of causes in the creation of all creatures, but not because this was His intention from the beginning. He never created man.

The philosopher is apparently comfortable with metaphor, at least in the phrasing of his own cosmology. But when the Rabbi has his turn, he rejects the philosopher's argument, and further argues that בראשית should be read as ironclad evidence against the philosopher's theory. This culminates in the following exchange:

אָמַר הַכּוּזָרִי: וְכִי זֶה מְחַיֵּב שֶׁלֹּא נַאֲמִין לַאֲרִיסְטוֹ בְּחָכְמָתוֹ.

אָמַר הֶחָבֵר: כֵּן, מִפְּנֵי שֶׁהִטְרִיחַ אֶת שִׂכְלוֹ וּמַחֲשַׁבְתּוֹ בָּעֲבוּר שֶׁלֹּא הָיְתָה בְיָדוֹ קַבָּלָה מִמִּי שֶׁיַּאֲמִין בְּהַגָּדָתוֹ [...] וְאִלּוּ הָיָה הַפִּילוֹסוֹף בְּאֻמָּה שֶׁיְּנַחֵל מְקֻבָּלוֹת וּמְפֹרָסָמוֹת שֶׁאֵינוֹ יָכוֹל לִדְחוֹת אוֹתָם, הָיָה מִתְעַסֵּק בְּהַקְשׁוֹתָיו וּמוֹפְתָיו לְהַחֲזִיק הַחִדּוּשׁ עִם קָשְׁיוֹ. בַּאֲשֶׁר הֶחֱזִיק הַקְּדָמוּת אֲשֶׁר הוּא יוֹתֵר קָשֶׁה לְקַבֵּל.

Al Khazari: Does this mean that Aristotle's philosophy is not deserving of credence?

The Rabbi: Certainly. He exerted his mind, because he had no tradition from any reliable source at his disposal. [...] Had he lived among a people with well authenticated and generally acknowledged traditions, he would have applied his deductions and arguments to establish the theory of creation, however difficult, instead of eternity, which is even much more difficult to accept.

[17] This just means "The Khazar," which is also the Hebrew title of the book (הכוזרי).

(Notice the claim here that a different understanding of tradition would have influenced the philosopher's cosmology—we'll return to that idea in the next section.) Halevi, via the character of the Rabbi, further argues against interweaving the textual narrative with a scientific perspective, and perhaps against studying cosmology empirically at all. He writes:

כִּי אֵין הַקַּבָּלָה טוֹבָה אֶלָּא עִם הַלֵּב הַטּוֹב, אֲבָל עִם רַע הַלֵּב הַחֲקִירָה יוֹתֵר טוֹבָה, כָּל שֶׁכֵּן כְּשֶׁיּוֹצִיא הַמֶּחְקָר אֶל אֱמוּנַת הַקַּבָּלָה הַהִיא, וְאָז יִתְקַבְּצוּ לָאָדָם שְׁתֵּי הַמַּדְרֵגוֹת, רְצוֹנִי לוֹמַר: הַיְדִיעָה וְהַקַּבָּלָה יָחַד.

Al Khazari: Tradition in itself is a good thing if it satisfies the soul, but a perturbed soul prefers research, especially if examination leads to the verification of tradition. Then knowledge and tradition become united.

אָמַר הֶחָבֵר: וּמִי לָנוּ בְּנֶפֶשׁ סוֹבֶלֶת שֶׁאֵינֶנָּה נִפְתֵּית לַדֵּעוֹת הָעוֹבְרוֹת עָלֶיהָ מִדֵּעוֹת הַטִּבְעִיִּים וְהָאַסְטַגְנִינִים וּבַעֲלֵי הַטַּלְסָמָאוֹת וְהַמְכַשְּׁפִים וְאַנְשֵׁי הַקֶּדֶמוּת וְהַפִּילוֹסוֹפִים וְזוּלָתָם, וְלֹא יַגִּיעַ אֶל הָאֱמוּנָה עַד שֶׁיַּעֲבֹר עַל דֵּעוֹת רַבּוֹת מֵהָאֶפִּיקוּרְסִים

The Rabbi: Where is the soul which is strong enough not to be deceived by the views of philosophers, scientists, astrologers, adepts, magicians, materialists, and others, and can adopt a belief without having first passed through many stages of heresy?

At this point, you may notice something interesting about the chronology: these texts originate from the 12th century CE, whereas Aristotle lived in the 4th century BCE. These discussions about Aristotelian cosmology are not direct responses to Aristotle. Rather, they reflect scientific discourse during the Islamic Golden Age, during which many classical Greek manuscripts reentered the zeitgeist. This era was among the first of several historical periods that elevated scientific cosmologies over traditional conceptions of the Universe. Another such period would later arrive with the European Enlightenment in the 18th century, and another with the rise of evolutionary theory in the 19th century, and yet another with the development of physical cosmology itself in the 20th century.

Each such era seems to have been associated with a flurry of scholarly activity among theologians. To my knowledge, most if not all instances of strict literalism in the reading of בראשית take the form of *responses* to the ascendance of scientific ideas which might have been taken as threats to the integrity of Jewish theology. In particular, I know of no instances where a major commentator analyzed an *internal* tension of the creation narrative, and insisted that we must not go beyond the plain meaning of the text in order to resolve the difficulty. We appear to have very little appetite for literalism in a vacuum.

Metaphorical interpretations of בראשית

Now, let us move on from strict literalism, and invoke metaphorical extensions of the text to put it in concordance with the scientific understanding of cosmology.

I want to be clear that I do not believe there is any point to doing this as an exercise in apologetics. Whether or not one can conceivably deform an ancient text to match a particular scientific narrative probably shouldn't have much influence on your regard for that text. After all, given sufficient metaphoric license, this can readily be done for *any* creation narrative. If you don't believe me, I encourage you to try it. (Fair warning: there are some raunchy creation myths out there.) So, the statement that one *can* do this for בראשית is vacuous. All

that interests me here is what one actually learns about the structure of the text in the course of making the attempt, and that is what I will try to highlight.

Let's first go through the text one day at a time, and then we can make more global statements. The narrative begins with the most basic elements of creation:

(א) בְּרֵאשִׁית בָּרָא אֱלֹקִים אֵת הַשָּׁמַיִם וְאֵת הָאָרֶץ: (ב) וְהָאָרֶץ
הָיְתָה תֹהוּ וָבֹהוּ וְחֹשֶׁךְ עַל־פְּנֵי תְהוֹם וְרוּחַ אֱלֹקִים מְרַחֶפֶת
עַל־פְּנֵי הַמָּיִם: (ג) וַיֹּאמֶר אֱלֹקִים יְהִי אוֹר וַיְהִי־אוֹר:
(ד) וַיַּרְא אֱלֹקִים אֶת־הָאוֹר כִּי־טוֹב וַיַּבְדֵּל אֱלֹקִים בֵּין הָאוֹר
וּבֵין הַחֹשֶׁךְ: (ה) וַיִּקְרָא אֱלֹקִים | לָאוֹר יוֹם וְלַחֹשֶׁךְ קָרָא לָיְלָה
וַיְהִי־עֶרֶב וַיְהִי־בֹקֶר יוֹם אֶחָד:

[1]In the beginning God created the heaven and the earth. [2]And the earth was without form and void; and darkness was on the face of the deep. And a wind from God moved over the surface of the waters. [3]And God said, Let there be light: and there was light. [4]And God saw the light, that it was good: and God divided the light from the darkness. [5]And God called the light Day, and the darkness He called Night. And there was evening and there was morning, one day.

If we were to put this in correspondence with the scientific chronology, what would these elements mean? It is difficult to even bound the time period that could correspond to the first day of creation. Interpreting the creation of "light" as the last physical event could put the end of the first day at electroweak symmetry breaking (section 6). Interpreting "light" more generally could even put this moment at reheating (section 4). On the other hand, if we interpret "day" and "night" in the context of the Earth's daily cycle, this last verse could take place after the formation of the Earth itself, almost ten billion years after the Big Bang (section 11).

That said, throughout part I, I expressed a few opinions on the most natural correspondences. A state of תהו ובהו (translated above as "without form and void"), although difficult to even understand, resembles the Big Bang plasma—i.e., the unstructured primordial Universe. Nachmanides writes of this state as follows:

הוציא מן האפס הגמור המוחלט יסוד דק מאד, אין בו
ממש, אבל הוא כוח ממציא, מוכן לקבל הצורה ולצאת
מן הכוח אל הפועל. והוא החומר הראשון, נקרא ליוונים
"היולי". ואחר ההיולי לא ברא דבר, אבל יצר ועשה [...]
והחומר הזה, שקראו היולי, נקרא בלשון הקדש "תֹּהוּ"...

He brought forth from total and absolute nothing a very thin substance devoid of corporeality but having a power of potency, fit to assume form and to proceed from potentiality into reality. This was the primary matter created by G-d; it is called by the Greeks *hyle* (ὕλη). After the hyle, He did not create anything, but He formed and made things with it [...] This substance, which the Greeks called hyle, is called in the sacred language *tohu* [תהו / "without form"]...

We will return to the Greek reference in the next section, but for the moment, I want to highlight Nachmanides' suggestion that תהו ובהו implies the first creation of the basic elements that can be formed into familiar matter. In the context of cosmology, this is clearly best aligned with reheating: the end of inflation, and the transferral of energy from the inflaton field to the ordinary matter fields, "fit to assume form" / מוכן לקבל הצורה.

What, then, do we make of the "wind" and "darkness" present in the same verse? I suggest that this fits best with the epoch of inflation itself (or whatever else might have preceded

reheating). In the inflationary paradigm, the structure of the Universe is essentially determined by the fields that drive inflation, whose behavior at these times is very far from that of corporeal matter. There is certainly no light in the Universe at this epoch, so "darkness" is appropriate. Does the "wind from God" (or, to translate it different, "spirit of God") describe a the rolling of a quantum field far from its vacuum? I can't say, but I also can't say how I could better describe such a phenomenon meaningfully in the language and style of בראשית. Then the creation of light corresponds to electroweak symmetry breaking, and the separation between light and darkness corresponds naturally to recombination, when light can finally propagate freely in the Universe. This is the first time at which light and darkness could have separately existed in the Universe.

This interpretation introduces a difficulty: inflation must come before reheating, but the "darkness" described by the text seems to come after the formless matter, to be ended only with the creation of light. That said, I think any honest attempt at constructing such a metaphorical correspondence must allow for the narrative to be given out of chronological order. We will revisit this point shortly.

An alternate interpretation is to consider the "wind" and "darkness" to be related to the expansion, and in particular, the *cooling* of the primordial Universe. In this interpretation, inflation (or whatever preceded the Big Bang) is omitted from the narrative entirely. Here, there is no chronological problem, because the cooling "wind" would have a causal relationship to the creation of light: electroweak symmetry breaking, remember, happens only once the Universe cools to a particular temperature. The same wind would then be the causative element in recombination and star formation. Broadly, though, both interpretations make an identification between divine substance and physical phenomena, à la Spinoza.

Let's continue with the second day.

(ו) וַיֹּאמֶר אֱלֹקִים יְהִי רָקִיעַ בְּתוֹךְ הַמָּיִם וִיהִי מַבְדִּיל בֵּין
מַיִם לָמָיִם: (ז) וַיַּעַשׂ אֱלֹקִים אֶת־הָרָקִיעַ וַיַּבְדֵּל בֵּין הַמַּיִם
אֲשֶׁר מִתַּחַת לָרָקִיעַ וּבֵין הַמַּיִם אֲשֶׁר מֵעַל לָרָקִיעַ וַיְהִי־כֵן:
(ח) וַיִּקְרָא אֱלֹקִים לָרָקִיעַ שָׁמָיִם וַיְהִי־עֶרֶב וַיְהִי־בֹקֶר יוֹם
שֵׁנִי:

[6]God said, "Let there be an expanse in the midst of the water, that it may separate water from water." [7]God made the expanse, and it separated the water which was below the expanse from the water which was above the expanse. And it was so. [8]God called the expanse Sky. And there was evening and there was morning, a second day.

In section 9, I argued that the "expanse" could be translated instead as "extended structure," and that such a structure could correspond with the formation of *cosmic* structures. Cosmic structures do form one dimension at a time: first, overdensities collapse into sheets, and sheets collapse into filaments, and filaments collapse into nodes (halos), separated by great voids. In this sense, cosmic structures really are what you get by thinly spreading the cosmic matter fluid. This interpretation forces us to make a correspondence between "water" and a more general fluid—but this is precisely the sort of abuse of language we commit in modern cosmology by calling everything a "fluid" on a technical basis.

The appeal I find in this interpretation really derives from the division between the fluid "below" (מתחת) and "above" (מעל) the structure. This is the first time that the narrative

exhibits a concept of *up* versus *down.* As proud residents of a spherical Earth, we know that up and down are not canonical, global directions, but instead determined locally by gravity. "Up" means "away from the center of mass," and "down" means "towards it." This only makes sense in a system that *has* a center of mass—i.e., a system that is not homogeneous. The collapse of the cosmological matter fluid into structures provides the first reference frames for the notion of up versus down. Thus, you can read this verse as actually indicating the creation of such distinct directions, and thus, the formation of the first substantial structures in the previously homogeneous Universe.

Let's continue with the third day.

(ט) וַיֹּאמֶר אֱלֹקִים יִקָּווּ הַמַּיִם מִתַּחַת הַשָּׁמַיִם אֶל־מָקוֹם אֶחָד
וְתֵרָאֶה הַיַּבָּשָׁה וַיְהִי־כֵן: (י) וַיִּקְרָא אֱלֹקִים ׀ לַיַּבָּשָׁה אֶרֶץ
וּלְמִקְוֵה הַמַּיִם קָרָא יַמִּים וַיַּרְא אֱלֹקִים כִּי־טוֹב: (יא) וַיֹּאמֶר
אֱלֹקִים תַּדְשֵׁא הָאָרֶץ דֶּשֶׁא עֵשֶׂב מַזְרִיעַ זֶרַע עֵץ פְּרִי עֹשֶׂה
פְּרִי לְמִינוֹ אֲשֶׁר זַרְעוֹ־בוֹ עַל־הָאָרֶץ וַיְהִי־כֵן: (יב) וַתּוֹצֵא
הָאָרֶץ דֶּשֶׁא עֵשֶׂב מַזְרִיעַ זֶרַע לְמִינֵהוּ וְעֵץ עֹשֶׂה־פְּרִי
אֲשֶׁר זַרְעוֹ־בוֹ לְמִינֵהוּ וַיַּרְא אֱלֹקִים כִּי־טוֹב: (יג) וַיְהִי־עֶרֶב
וַיְהִי־בֹקֶר יוֹם שְׁלִישִׁי:

[9]God said, "Let the water below the sky be gathered into one area, that the dry land may appear." And it was so. [10]God called the dry land Earth and called the gathering of waters Seas. And God saw that this was good. [11]And God said, "Let the earth sprout vegetation: seed-bearing plants, fruit trees of every kind on earth that bear fruit with the seed in it." And it was so. [12]The earth brought forth vegetation: seed-bearing plants of every kind, and trees of every kind bearing fruit with the seed in it. And God saw that this was good. [13]And there was evening and there was morning, a third day.

This day is extremely problematic. There are major problems with a literal reading, but let's begin by acknowledging a problem with the metaphor that we have so far established: here, "water" clearly means ordinary water, and is not a stand-in for a generic cosmological fluid. Should that trouble us?

I would argue that it should not trouble us any more than the previous references to water, which clearly must *not* refer to ordinary water in a semi-literal reading. That is, you cannot argue that "water" (מים) always means ordinary water, and simultaneously hold that it was present before the creation of light! Thus, I suggest interpreting "water" as a stand-in for fluidlike *behavior,* in the same way that cosmologists use the term "fluid." We only describe the matter fluid as a "fluid" at early times and on large scales, where this description works beautifully. But neutrinos, for example, are importantly *not* a perfect fluid at late times. Their effective behavior on large scales is different, even though the neutrinos themselves have not changed at all. If we read "water" as "fluid," then it's entirely appropriate that the definition of "water" changes with scale, just as does the definition of "fluid." Solid matter is fluidlike on large scales, in the cosmological context, but it is not fluidlike on the length scales and time scales relevant to us on Earth. This means that even though the Earth itself is part of the cosmological matter fluid on sufficiently large scales, it is still useful to speak of separating between solids and fluids on Earth.

Now, there are three other issues that any metaphoric interpretation should explain.

1. The text says that the water gathered and receded, giving rise to dry land. What does this correspond to in the Earth's history?

2. The creation of plants is out of order. Even if we ignore the fact that animals haven't been created yet, plants with seeds only evolved about 300 million years ago. And what of all the plants without seeds?

3. Plants are created before the Sun! How could they grow?

The first point is the easiest: here we can make a direct correspondence with the conditions of the Earth's surface during the Hadean era, in the first few hundred million years after its formation. Liquid water quickly appeared despite the high temperatures, and likely covered most if not all of the Earth's surface. The water level fluctuated substantially in the epochs that followed, so the text could plausibly be referring to this migration of water, which did give rise to continents. But for the second point, I cannot give any better interpretation than rejecting the premise of chronological order in the creation narrative. We will revisit the last point after we finish reading through the story.

Let's continue with the fourth day.

(יד) וַיֹּאמֶר אֱלֹקִים יְהִי מְאֹרֹת בִּרְקִיעַ הַשָּׁמַיִם לְהַבְדִּיל
בֵּין הַיּוֹם וּבֵין הַלָּיְלָה וְהָיוּ לְאֹתֹת וּלְמוֹעֲדִים וּלְיָמִים
וְשָׁנִים: (טו) וְהָיוּ לִמְאוֹרֹת בִּרְקִיעַ הַשָּׁמַיִם לְהָאִיר עַל־הָאָרֶץ
וַיְהִי־כֵן: (טז) וַיַּעַשׂ אֱלֹקִים אֶת־שְׁנֵי הַמְּאֹרֹת הַגְּדֹלִים
אֶת־הַמָּאוֹר הַגָּדֹל לְמֶמְשֶׁלֶת הַיּוֹם וְאֶת־הַמָּאוֹר הַקָּטֹן
לְמֶמְשֶׁלֶת הַלָּיְלָה וְאֶת הַכּוֹכָבִים: (יז) וַיִּתֵּן אֹתָם אֱלֹקִים
בִּרְקִיעַ הַשָּׁמָיִם לְהָאִיר עַל־הָאָרֶץ: (יח) וְלִמְשֹׁל בַּיּוֹם וּבַלַּיְלָה
וּלְהַבְדִּיל בֵּין הָאוֹר וּבֵין הַחֹשֶׁךְ וַיַּרְא אֱלֹקִים כִּי־טוֹב:
(יט) וַיְהִי־עֶרֶב וַיְהִי־בֹקֶר יוֹם רְבִיעִי:

[14]God said, "Let there be lights in the expanse of the sky to separate day from night; they shall serve as signs for the set times—the days and the years; [15]and they shall serve as lights in the expanse of the sky to shine upon the earth." And it was so. [16]God made the two great lights, the greater light to dominate the day and the lesser light to dominate the night, and the stars. [17]And God set them in the expanse of the sky to shine upon the earth, [18]to dominate the day and the night, and to separate light from darkness. And God saw that this was good. [19]And there was evening and there was morning, a fourth day.

The Sun is apparently created after the Earth, after the creation of plants, and simultaneously with the Moon. The Sun did exist when the Earth first formed, but its behavior then was quite different from today. In particular, for the first 10–100 million years after Earth formed, the Sun was a T Tauri star, a kind of young star powered by the energy of infalling gas rather than by nuclear fusion. So, perhaps the creation of the Sun refers to its start on the stellar "main sequence," i.e., the evolutionary trajectory of a broad class of stars powered by fusion. In this picture, it's appropriate that the Sun and Moon are "created" shortly after the Earth.

Let's finish with the fifth and sixth days.

(כ) וַיֹּאמֶר אֱלֹקִים יִשְׁרְצוּ הַמַּיִם שֶׁרֶץ נֶפֶשׁ חַיָּה וְעוֹף
יְעוֹפֵף עַל־הָאָרֶץ עַל־פְּנֵי רְקִיעַ הַשָּׁמָיִם: (כא) וַיִּבְרָא אֱלֹקִים
אֶת־הַתַּנִּינִם הַגְּדֹלִים וְאֵת כָּל־נֶפֶשׁ הַחַיָּה | הָרֹמֶשֶׂת אֲשֶׁר
שָׁרְצוּ הַמַּיִם לְמִינֵהֶם וְאֵת כָּל־עוֹף כָּנָף לְמִינֵהוּ וַיַּרְא
אֱלֹקִים כִּי־טוֹב: (כב) וַיְבָרֶךְ אֹתָם אֱלֹקִים לֵאמֹר פְּרוּ וּרְבוּ
וּמִלְאוּ אֶת־הַמַּיִם בַּיַּמִּים וְהָעוֹף יֵרֶב בָּאָרֶץ: (כג) וַיְהִי־עֶרֶב
וַיְהִי־בֹקֶר יוֹם חֲמִישִׁי:

[20]God said, "Let the waters bring forth swarms of living creatures, and birds that fly above the earth across the expanse of the sky." [21]God created the great sea monsters, and all the living creatures of every kind that creep, which the waters brought forth in swarms, and all the winged birds of every kind. And God saw that this was good. [22]God blessed them, saying, "Be fertile and increase, fill the waters in the seas, and let the birds increase on the earth." [23]And there was evening and there was morning, a fifth day.

(כד) וַיֹּאמֶר אֱלֹקִים תּוֹצֵא הָאָרֶץ נֶפֶשׁ חַיָּה לְמִינָהּ בְּהֵמָה
וָרֶמֶשׂ וְחַיְתוֹ־אֶרֶץ לְמִינָהּ וַיְהִי־כֵן: (כה) וַיַּעַשׂ אֱלֹקִים
אֶת־חַיַּת הָאָרֶץ לְמִינָהּ וְאֶת־הַבְּהֵמָה לְמִינָהּ וְאֵת כָּל־רֶמֶשׂ
הָאֲדָמָה לְמִינֵהוּ וַיַּרְא אֱלֹקִים כִּי־טוֹב: (כו) וַיֹּאמֶר אֱלֹקִים
נַעֲשֶׂה אָדָם בְּצַלְמֵנוּ כִּדְמוּתֵנוּ וְיִרְדּוּ בִדְגַת הַיָּם וּבְעוֹף
הַשָּׁמַיִם וּבַבְּהֵמָה וּבְכָל־הָאָרֶץ וּבְכָל־הָרֶמֶשׂ הָרֹמֵשׂ
עַל־הָאָרֶץ: (כז) וַיִּבְרָא אֱלֹקִים | אֶת־הָאָדָם בְּצַלְמוֹ בְּצֶלֶם
אֱלֹקִים בָּרָא אֹתוֹ זָכָר וּנְקֵבָה בָּרָא אֹתָם:

[24]God said, "Let the earth bring forth every kind of living creature: cattle, creeping things, and wild beasts of every kind." And it was so. [25]God made wild beasts of every kind and cattle of every kind, and all kinds of creeping things of the earth. And God saw that this was good. [26]And God said, "Let us make humankind in our image, after our likeness. They shall rule the fish of the sea, the birds of the sky, the cattle, the whole earth, and all the creeping things that creep on earth." [27]And God created humankind in the divine image, creating it in the image of God—creating them male and female.

Perhaps the biggest problem in these final two days of creation is the temporal separation between different categories of animals. This tension can be slightly ameliorated by viewing this organization as a nod to the oceanic origin of life, although such a resolution does nothing to address the yet earlier creation of plants. Furthermore, birds specifically evolved from land animals, so if the narrative is hinting at the evolutionary history of life, it makes no sense for birds to have been created before land animals.

You could try to make chronological sense of this with a very generous interpretation: that the fifth day describes previous eras of life on Earth whose representatives are now extinct. One hint in favor of this reading is the mention of תנינם, which falls a bit flat in translation as "great sea monsters." The word תנין has been variously interpreted as "dragon" or "serpent," and in modern Hebrew, it means "crocodile" or "alligator." If this refers to marine megafauna, then one could interpret likewise "birds" as the extinct dinosaurian relatives of modern birds, which are the only remaining dinosaurs. But this would be an aggressive interpretation: although such megafauna tends to grab our imagination of ancient Earth, this really isn't a good description of the history of life on our planet.

So, having now surveyed all six days of creation, it is time to reckon with the issue of chonological order.

Plants without the Sun

The order of creation of animals is only the last example of several problems we have noted with a chronological reading. Crucially, one does not need an understanding of modern physical cosmology nor evolutionary biology to observe such chronological problems as the creation of plants before Sun. These problems were apparent to many generations of commentators, and some traditional sources directly endorse the idea of a nonchronological narrative. In the era before science and religion clashed, this was not seen as a third rail! Even Rashi, one of the more literally-minded commentators, wrote regarding an apparent chronological issue in Genesis 6:3 / בראשית ו׳:ג׳ as follows:

וְאִם תֹּאמַר מִשֶּׁנּוֹלַד יֶפֶת עַד הַמַּבּוּל אֵינוֹ אֶלָּא מֵאָה שָׁנָה,
אֵין מֻקְדָּם וּמְאֻחָר בַּתּוֹרָה

And if you say that from the birth of Japheth until the Flood there were only 100 years, [which would imply a chronological conflict, you must remember that] there is no "earlier" or "later" in the Torah.

Rashi's principle, אין מוקדם ומאוחר בתורה, or "There is no 'earlier' and 'later' in the Torah," has since evolved into a frequently-used principle in Jewish theology: that the narrative can be viewed as lying outside of time, such that it sometimes presents events out of our chronological order in order to teach some particular lesson. Rashi himself used this principle often. It has since been applied by many commentators to a variety of chronological challenges. That is not to say that it has been universally accepted, but it has certainly been considered a reliable principle by many traditional sources. It seems to me that we must accept this principle if we are to make sense of the narrative regarding the origin of life. We can wriggle out of some difficulties as I sketched above, but we can't explain the creation of fruiting trees before the first animals without abandoning a strict interpretation of the chronology.

If the creation of animals is presented out of chronological order in order to teach us something, what is that lesson? I don't have a great answer. I do note that this is the first time that תנ״ך presents a taxonomy of life, and the separation of days serves to establish distinct categories. These taxonomic categories are later relevant to issues of ritual purity and kashrut / בשרות. Perhaps that is reason enough to make these groupings, even as they are orthogonal to the literal order in which life arose on Earth.

An even more challenging problem is the existence of plant life prior to the Sun and stars. Perhaps recognizing the issue, Nachmanides writes:

He decreed that there be among the products of the earth a force which grows and bears seed so that the species should exist forever.	גזר שיהיה בתולדות הארץ כח הצומח ומוליד זרע כדי שיהיה המין קיים לעד

If we interpret the action on this day as the creation of potential for life, rather than life itself, that seems at first to ease the tension slightly. However, our modern understanding of physical cosmology makes this untenable, for the following reason: *life on Earth requires carbon.* Carbon, being heavier than Lithium, is not produced in any appreciable quantity in the Big Bang. It is only produced by stellar nucleosynthesis much later on. Plants—and indeed, the rocky Earth—cannot exist prior to the deaths of the first generation of stars. For this reason, I think that a strictly chronological reading is untenable, no matter how long a "day" is taken to be.

How long is a day?

Perhaps the most obvious problem in the entire creation narrative is the notion of "days." There are two problems: first, how can one reconcile the duration of a single day with the corresponding timescales for cosmological processes that took fractions or a second or millions of years? Second, what even *is* a "day," let alone a "evening" and a "morning," prior to the creation of the Sun? There are three possible resolutions to the first problem:

1. The Universe was created in a literal set of six 24-hour periods, but at each stage, features were created with the appearance of age compatible with scientific chronology.

2. Six literal "days" can be made compatible with the timescales of physical cosmology.

3. The use of "day" is metaphorical and stands in for an epoch, organizing phenomenological periods rather than uniform time periods.

It will not surprise you to know that I prefer the third. Let's discuss the other possibilities.

The first resolution is called the "Omphalos hypothesis," from the Greek word for "navel," since it posits that Adam had a bellybutton even though he never had an umbilical cord. Is it possible? Certainly—if we fundamentally reject the possibility of studying cosmic history empirically, then we could say anything. I find this deeply troubling. It implies a deceptive creator who foresaw our eventual abilities to look into the past, and seeded this past with evidence of events that never took place. This seems to me to be absurd. What possible reason would a divine creator have to create such a lie simply for the purpose of justifying a literal interpretation of the word "day" in a human language? Moreover, if the creator of the Universe is happy to deceive us, why should we trust that the Torah itself is free of deception?

A further argument is the *reductio ad absurdum* of the Omphalos hypothesis: if one accepts that the Universe was created with evidence of a past that never happened, why stop at creation six thousand years ago? The Universe may as well have been created six *minutes* ago, complete with our understanding of Torah. This is sometimes called *Last Thursdayism* in parody of unfalsifiable religious claims: a religious sect of Last Thursdayists could insist that the Universe was created last Thursday, and this would be impossible to rebut.

What about the second resolution? Here, the argument is that the physical flow of time was different in some way, such that a 24-hour period literally meant something different in each day of creation. Now, we have a very good theoretical framework for treating the rate of passage of time according to different observers: this is exactly what general relativity allows us to do. Some authors have tried to take advantage of the relativity of time to effect such a changing "day," notably Gerald Schroeder.

Regrettably, this is simply not viable, which you might have suspected from the simple fact that we cosmologists are very much aware of relativity when we compute the age of the Universe! In fact, when we say that 13.7 billion years has elapsed since the Big Bang, we are actually speaking very precisely about the time as measured by an observer in a "comoving reference frame." You could certainly imagine a (noninertial) reference frame in which an observer measures a very different age, and even different cosmological epochs for different "days," but this is vacuous. The lesson of relativity is that a time specified without a reference frame is meaningless. If these time periods are secretly dependent on an abstruse choice of reference frame, why specify them at all, let alone as days? This, too, is absurd.

A further argument against interpreting the days in terms of exact 24-hour periods is that the length of a day is not actually uniform! Since the Earth's orbit around the sun is not perfectly circular, its rate of rotation around the Sun varies ever so slightly over the course of a year. Since the apparent motion of the Sun is actually a combination of Earth's rotation about the Sun and its rotation about its own (tilted) axis, this means that the length of a day actually varies by about 30 seconds. This is before we consider the fact that the Solar System is constantly evolving, and the length of a day increases by about 1.7 milliseconds

every century. Therefore, even on astronomical grounds, we must accept that the chronology given in בראשית is *inherently* imprecise.

Given these arguments, I personally have difficulty with any interpretation that tries to make the six-day time period literal in any fashion. This is simply incompatible with our understanding of the Universe. One is of course free to jettison scientific principles, e.g. by employing the Omphalos hypothesis, but I find that to be a much harder pill to swallow than a metaphorical interpretation. I will argue for an even broader role of metaphor in section 17.

14 The literary reading: creation in an ontological context

In the previous section, we discussed the boundaries of literal interpretation, and I argued that strict literalism is untenable. But there is an important part of the mechanics of a literal reading that deserves separate attention: the actual meaning that we ascribe to the language of creation. In this section, I want to question the relationship between the actual *words* used to describe the formation of the natural world in each of several contexts: the Hebrew language of the textual narrative, ancient philosophical traditions in the languages of other cultures, and modern physical cosmology in the mathematized language of physics. How do the modes of expression compare, and how does this influence the nature of dialectic between traditional sources and modern science? I call this form of inquiry the *literary* reading.

I want to make a distinction between the question I am posing here and what one might ask in the framework of comparative religious studies. I am *not* interested here in reading בראשית as a creation myth that inherits from other cultures. This is certainly an interesting line of study, and a great deal has been written and said on this subject. However, I will not touch this, for two reasons. Firstly, I am simply not an expert on any part of the subject. Secondly, this reading deviates from the position of theological textual analysis that I want to assume for the purposes of these notes. In contrast to most of my life as a scientist, I am interested here in reading the narrative as a religious text, not as a historical artifact. There is some suspension of disbelief in this act for me, and perhaps for you too. I accept that as a premise for the purpose of engaging in the exercise of Jewish textual study, in the same way that one might accept an unusual set of axioms for the purpose of a particular mathematical exercise. The bottom line is that I am not presently allowing for features of בראשית to be explained by human imagination or frailty.

Now, with that said, here is the premise of the current question. The textual narrative is clearly answering a set of questions about the nature of the Universe. Those questions are selected according to the *ontology* of the presumed audience, i.e., their conceptual categories of what kinds of objects exist in which fashions. Modern physical cosmology has a dramatically different ontology, which leads to a different set of questions with incommensurable answers. For the most basic example, consider the starting point that we adopted in section 2: the large-scale behavior of the Universe is phrased in terms of the structure of spacetime, a construct that endows our dimensions of daily experience with geometry and topology. Such an association would be alien to a reader who had never encountered non-Euclidean geometry,[18] let alone general relativity.

In particular, consider the following entities: space, time, substance, emptiness, and energy; and the Hebrew words ראשית ["beginning"], שמים ["sky"], ארץ ["land"], תהו ובהו ["without form and void"], יום ["day"], יהי ["let there be"], אור ["light"], and טוב ["good"]. We have

[18]The structure of spacetime is modeled on pseudo-Riemannian geometry, which is non-Euclidean. I emphasize the deviation from *Euclidean* geometry because that was the cultural basis for mathematical reasoning until the 18th century, and certainly represented a basic touchstone of scientific logic in the era of the traditional commentators. The rise of non-Eucldean geometry was an ontological shift even among mathematicians.

already touched on interpretations of many of these words in the preceding sections. These are all nontrivial concepts that have embedded cultural presumptions, even in their scientific incarnations. Indeed, we will see shortly that even the conceptual ontology of modern physical cosmology is not necessarily free of the fingerprints of its cultural antecedents.[19]

The creation narrative itself does not supply us with a metaphysical philosophy, which is part of what complicates the intepretation of the text today. So, what does the following line mean?

(א) בְּרֵאשִׁית בָּרָא אֱלֹקִים אֵת הַשָּׁמַיִם וְאֵת הָאָרֶץ: (ב) וְהָאָרֶץ
הָיְתָה תֹהוּ וָבֹהוּ וְחֹשֶׁךְ עַל־פְּנֵי תְהוֹם וְרוּחַ אֱלֹקִים מְרַחֶפֶת
עַל־פְּנֵי הַמָּיִם:

[1]In the beginning God created the heaven and the earth. [2]And the earth was without form and void; and darkness was on the face of the deep. And a wind from God moved over the surface of the waters.

We have already discussed metaphorical interpretations of these terms in section 13. Now let's take a literary eye to the same verses. The first thing to point out that is that God *created in* the beginning—that is, God does not *create the* beginning. The idea that one could create *time* relies on having a notion of time as a physical entity. That is something that exists in our relativistic ontology, but likely had no meaning in the narrative ontology. The Midrash, in בראשית רבה, goes so far as to identify the object of "beginning" with the Torah itself:

אָמְרָה תּוֹרָה בְּרֵאשִׁית בָּרָא אֱלֹהִים. וְאֵין רֵאשִׁית אֶלָּא תּוֹרָה

The Torah says: "In the beginning God created," and "beginning" means nothing other than the Torah itself.

It is worth mentioning here that Torah / תורה is itself an ontological entity in Jewish thought that goes well beyond a particular book or corpus of text. The word is sometimes translated literally as "teaching," but it might be better translated as "deep truth." The Rabbis go so far as to say that Torah is one of the underlying forces sustaining the existence of the Universe. So, the suggestion that God created Torah denotes the origin of a different category which, like time, seems at first to be inseparable from existence.

This hints at the nature of the question being answered in the opening verses of בראשית. Both the textual narrative and physical cosmology come up just to the *edge* of cosmogony. In the case of physical cosmology, the question—which remains unanswered—regards the origin of the entities of space and time. In the narrative, suggests the Midrash, the starting question regards the origin of the very notion of orderly truth. But it clearly has no bearing on the origin of *time* as an entity. Indeed, in a straightforward reading, time exists prior to the narrative of creation.

After all, what would it mean to create time? Creation is a causal activity: it is an action which precedes and conditions a consequence. In the absence of time, there is no causality. In the physical ontology, we can at least somewhat answer this question by differentiating between two notions of time, one microscopic and one statistical. The former is nearly fully

[19]This should not be misconstrued to imply that modern cosmology is in any way *incorrect.* Cosmology, like any scientific field, is judged on the basis of its predictions and observables.

reversible, whereas the latter has an entropic arrow of time. In this sense, the "creation" of time could be interpreted as the formation of a low-entropy state that would give rise to a consistent time direction (see section 12). At least in principle, this could take place by an unlikely coincidence of microphysical evolution, without requiring a consistent orientation to time in the pre-temporal state. But in the narrative ontology, there is no secondary, nonstatistical notion of time! There is thus no entity that could accommodate the causal action of a creator *prior* to time.

As a consequence, I argue that it is *intrinsic to premathematical language* that the creation narrative does not account for the origin of time. There is simply no way to logically describe the origin of time without first developing a statistical ontology.

This does not mean that the creation narrative should be read to imply that God did not create time. It simply indicates that the language of the narrative is expressively incapable of describing such a creation. For example, Maimonides, in *Guide for the Perplexed,* suggests that there was no time before creation:

> We find that some of our Sages are reported to have held the opinion that time existed before the Creation. But this report is very doubtful […] Those who have made this assertion have been led to it by a saying of one of our Sages in reference to the terms "one day," "a second day." Taking these terms literally, the author of that saying asked, What determined "the first day," since there was no rotating sphere, and no sun?
>
> אבל מה שתמצאהו בתוב לקצת ה׳חכמים׳ מהעמיד זמן נמצא קודם בריאת העולם – מסופק מאד […] ואשר הביא האומרים אל זה המאמר הוא מצאם: ״יום אחד״ ו״יום שני״ – והבין אומר זה המאמר את הענין על פשוטו וחשב שאחר שלא היה שם גלגל סובב ולא שמש באי זה דבר שוער ׳יום ראשון׳?

Setting aside the fact that Maimonides directly participates in the conversation over the nature of a "day" (section 13), he comes up to the boundary of describing the origin of time *by negation,* i.e., by saying that there was no time *before.* But even in this retrospective analysis, Maimonides cannot speak of time as an ontological entity that had been created, and thus cannot attribute its creation to God.

I wish to jump directly from "time" to "energy." This might seem a bit out of place, but I assure you it is not. Time provides the means by which we define energy in physics.[20] As such, an origin for the entity of time provides an origin for the entity of energy. I have mentioned several times in part I that energy is not truly conserved in cosmology. A more precise statement would be that energy is locally conserved, but cannot be globally *defined,* and thus cannot be said to be globally conserved. The local conservation of energy—or, equivalently, the local invariance of the laws of physics with time—is an extremely useful principle that has thus infiltrated our ontology of the Universe. We talk about energy in the Universe as though it fundamentally exists, even as cosmologists, while all the while we know that only energy density can be meaningfully defined.

One of the key features of energy in our physical ontology is its relation to matter. We have an intuitive notion of matter—"substance"—and providing for an origin to energy gives

[20] Energy is just the Noetherian conserved quantity associated with time translation symmetry.

rise to an origin for the matter of everyday experience. But this is an extremely modern notion, only about a century old. Substance has been the object of metaphysical discussion for thousands of years. It is thus informative to study the role of substance in narrative and rabbinic ontologies. The latter drew heavily on the Greek philosophical tradition, which developed an ontology of substance that dominated scientific discourse for many centuries.

We have already seen one rabbinic allusion to the Greek conception of substance, by Nachmanides, who equated תהו / "without-form" with ὕλη ("hyle"), the Greek notion of basic matter. Maimonides, in the *Guide*, also drew a correspondence between textual categories and the Greek entities of substance, i.e., the elements:

The four elements indicated, according to our explanation, in the term "earth," in the first verse, are mentioned first after the heavens: for there are named erez̩ (earth), ruaḥ (air), mayim (water), and hoshek (fire).

וממה שצריך שתדעהו – כי היסודות הארבע אשר אמרנו ששם ׳ארץ׳ הראשון יורה עליהם נזכרו תחלה אחר השמים. שהוא זכר ׳ארץ׳ הראשון יורה עליהם נזכרו תחילה אחר השמים. שהוא זכר ׳ארץ׳ ו׳מים׳ ו׳רוח׳ ו׳חושך׳

Given these correspondences, let us examine the classical Greek ontology of substance. This is largely established in Aristotle's *Metaphysics.* The *hyle* that Nachmanides refers to is introduced as follows:

Natural generation is the generation of things whose generation is by nature. That from which they are generated is what we call matter (ὕλην); that by which, is something which exists naturally; and that which they become is a man or a plant or something else of this kind, which we call substance in the highest degree. All things which are generated naturally or artificially have matter; for it is possible for each one of them both to be and not to be, and this possibility is the matter in each individual thing.

αἱ δὲ γενέσεις αἱ μὲν φυσικαὶ αὗταί εἰσιν ὧν ἡ γένεσις ἐκ φύσεώς ἐστιν, τὸ δ᾽ ἐξ οὗ γίγνεται, ἣν λέγομεν ὕλην, τὸ δὲ ὑφ᾽ οὗ τῶν φύσει τι ὄντων, τὸ δὲ τὶ ἄνθρωπος ἢ φυτὸν ἢ ἄλλο τι τῶν τοιούτων, ἃ δὴ μάλιστα λέγομεν οὐσίας εἶναι—ἅπαντα δὲ τὰ γιγνόμενα ἢ φύσει ἢ τέχνῃ ἔχει ὕλην: δυνατὸν γὰρ καὶ εἶναι καὶ μὴ εἶναι ἕκαστον αὐτῶν, τοῦτο δ᾽ ἐστὶν ἡ ἐν ἑκάστῳ ὕλη.

To appreciate what this means, we need to briefly recount the notion of *forms.* Plato, Aristotle's teacher, had constructed a philosophical doctrine that separated actual things from ideal versions of those things, which he called forms. For every universalizable entity, say, "table", Plato posited that there existed a corresponding form, say, the "form of table." This form would represent the most perfect embodiment of tablehood. Then a particular table in the real world would be considered to *be* a table insofar as it participated in the "form of table." But a particular table in the real world would only imperfectly participate in the form of table, which would represent the *Platonic ideal* of this ontological category. Plato specifically insisted on the realism of the form even in the absence of any particular exemplar thereof.

Aristotle disagreed with his teacher. He articulated a theory of *hylomorphism,* i.e., one predicated on a mixture of ὕλη ("hyle," matter) and μορφή ("morphe", form). Here, Aristotle posited the independent existence of matter without form, i.e., a medium that lacked any of the attributes that would make it a particular thing. He then asserted that the forms

were only realized in combination with matter. Matter is that which has capacity to become something—*potentiality*—and the *substantial form* is what turns this potentiality into *actuality* in the existence of a particular thing. This is what Aristotle means when he says that the possibility "to be and not to be," i.e., the potentiality for particularized existence, "is the matter in" a thing.

Aristotle identified the hyle in physical objects with the four elements: earth, air, water, and fire. In pointing to the creation of these four elements in בראשית, Maimonides was probably engaged in exactly the same exercise we undertook in section 13, attempting to map the textual narrative onto a contemporary scientific ontology. Interestingly, just as this is a difficult exercise today, it apparently required suspension of disbelief even in Maimonides' day: I know of nobody else who would translate חושך as "fire" instead of "darkness," its plain meaning, which is almost the opposite. Still, the fact that this sort of comparison has been a part of the intellectual tradition of the last two thousand years is evident even from the fact that תהו is translated as "without form"—*form.* This has meaning to the translator, and to us, only because we have some cultural notion of form, whose linguistic relation to the Platonic concept is no accident.

But Aristotle's hylomorphism was not the last step in the metaphysics of universals and particulars. Indeed, today, this seems like a much less important area of study. We tend to interpret the natural world through the lens of materialist reductionism. If everything is made of atoms and molecules of particular types bonded in particular ways, what need have we of forms? Thus, when I attempted a metaphorical mapping in section 13, I argued that תהו could be analogous to the undifferentiated Big Bang plasma. This is just a modern version of the same correspondence, projected onto a more modern physical ontology.

The continuation of this discussion could fill volumes. We haven't even begun to analyze most of the terms that I listed at the beginning of this section. For the moment, though, I want to summarize what I have said so far and articulate one more claim.

The argument above has one central thesis: that בראשית reads the way it does because it is written in a particular language, directed at a particular culture, anchored in a particular ontology. Accordingly, as our collective ontology for cosmic phenomena evolves, we need to fill more and more gaps in order to map the narrative onto our conceptual framework. In this picture, the difficulty in drawing such correspondences is not because of a failing of the text, but an inherent incapacity of its medium, which we are slowly and collectively remedying by extending this capacity over time. Crucially, I argue that this makes scientific interpretation of the text all the more worthwhile: as I demonstrated with the example of time itself, we eventually develop new ontological categories that give us an intellectual framework to pose deeper questions and understand deeper principles of the story of creation.

Before I close this section, I want to point out one more thing: the scientific conceptualization of cosmology is not free from the same ontological considerations. In some ways, this is obvious. As I argued with the example of general relativity, nobody who lived a thousand years ago would have framed their understanding of the cosmos in the same language that we use, and this is the natural course of evolution of science. But I want to go somewhat further

than that. I would argue that we often describe cosmology in ontological categories that we *know* are probably not fundamental to nature, simply because they are more familiar to us.

One good example of this is again general relativity. We are very accustomed to saying that "spacetime is curved." Maybe it is. But the general theory of relativity only accommodates a geometric description because of the *equivalence principle,* i.e., the statement that everything is equally accelerated by gravity. But the equivalence principle may well be only approximate. In candidate quantum theories of gravity, which would eventually replace general relativity, there are generically additional degrees of freedom that lead to very small deviations from the equivalence principle. These small deviations have no pragmatic importance for the application of general relativity to many physical systems, but they do have ontological significance for the interpretation of gravitation in terms of spacetime curvature. The geometric description only works because the equivalence principle is assumed to be *perfectly* satisfied. If it were not, we probably would never have phrased gravitation in such terms. Now that we know it is very plausible that the equivalence principle is in fact violated, that should affect our ontological interpretation of gravitation—we should not say that spacetime is truly curved, but only that such a mathematical model is very effective at predicting behavior on the scales we can currently test.

This example is again obvious in some way, and it has a clear resolution. We don't need to think of gravity in terms of geometry to do computations, but it's an effective conceptual tool, and we can be adult enough to separate the model from the phenomenon. What I find more troubling is that hidden versions of the same conceptual laziness may be lurking in many other places in our modern physical ontology. For example, the very idea of a field reeks of conceptual notions like the infinite divisibility of space and nature's abhorrence of a vacuum that each originated long before they could be put on any real empirical foundation. As such, while I know how quantitatively *right* we are, I can only guess at how ontologically *wrong* we are.

In making the above statements, I want to be extremely clear about three points. First, this has absolutely no bearing on the success of cosmological theory. The success of any scientific theory isn't judged on its ontological tenets, but on its predictive capacity, which is exactly why one ought to take its ontological interpretations with a grain of salt. Second, I don't think that this reflects any shortsightedness on the part of working physicists. If anything, in my experience, physicists are the *most* likely people to question these kinds of assumptions about the world. Third, I am not advocating for a reinvention or reconsideration of our physical ontology. To do so in a vacuum would probably be counterproductive. History suggests that we usually adapt very capably and very rapidly once empirical evidence becomes available to suggest and support new ontological interpretations. The way to a more perfect ontology of the Universe probably doesn't lie in deeply thinking about it so much as in continuing to advance and incrementally *break* our mathematical models of the Universe. That is how we advance our collective ontology. Just be patient and give us some time.

15 Mystical interpretations (קבלה)

A third approach to the joint interpretation of narrative and physical cosmologies is mystical, or *kabbalistic.* Mysticism is a notoriously slippery concept. It is founded on esoteric spiritual practices that are experiential by their nature. However, I nonetheless want to make some effort to engage with it here because of its relationship with creation in Jewish thought. The Jewish relationship with creation acknowledges a kind of secrecy or mystery about the exact circumstances of creation, as though it is founded on ideas that cannot be understood by ordinary cognitive processes. (This is not so different from what I argued in the previous section.) The mystical response to that state of affairs is to attempt a different sort of cognitive process, one that fosters an ecstatic experience of a realm that cannot be understood otherwise. Kabbalistic writings are the projections of these interpretive experiences onto the inadequate basis of language.

Personally, I find mysticism very difficult to understand. I have never had something that I would call a mystical experience. But as this is one of the most significant intellectual traditions in the Jewish understanding of creation, it is nonetheless valuable to ask what intersection mysticism has with our understanding of the physical Universe. This is not at all a question that should degrade the mystical experience. After all, Kabbalah / קבלה is ultimately about receiving insight about the interaction between an infinite, perfect God and a finite, imperfect world. How could this be carried out but with a firm grounding in the latter? However, much as I would like to represent the mystical viewpoint, I have virtually no knowledge of kabbalistic literature. Most of what is written here is borrowed from Howard Smith's book *Let There Be Light: Modern Cosmology and Kabbalah.*

The organizing principles of קבלה are the ten ספירות (*Sefirot*). These are ten forms of manifestation of the infinite, ineffable version of the divine prior to any creation, which is entirely universal and participates in no particulars. This idealized version of divinity is called אין סוף (*Eyn Sof*, "[that which has] no end" / "infinity"). The connection between אין סוף and the finite, particularized world—including that connection by which the latter is sustained in existence—is via the ספירות. Namely, these are (1) כתר (*Keter,* "crown"), (2) דעת (*Da'at*, "knowledge"), (3) חכמה (*Chokhmah*, "wisdom"), (4) בינה (*Binah*, "understanding"), (5) חסד (*Chesed*, "loving-kindness"), (6) גבורה (*Gevurah*, "strength"), (7) תפארת (*Tiferet*, "beauty" / "glory"), (8) נצח (*Netzach*, "perpetuity"), (9) הוד (*Hod*, "majesty" / "splendor"), (10) יסוד (*Yesod*, "foundation"), and (11) מלכות (*Malkhut*, "kingship"). Yes, there are eleven principles, not ten, but כתר and דעת are identified to yield ten (or the latter is omitted from the count entirely). The ספירות are further associated to varying degrees with four עולמות (*Olamot*, "worlds"), representing distinct planes of existence. Each of the עולמות is dominated by one or more of the ספירות, in the following order:

1. עצילות (*Atzilut*, "emanation"): This world is closest to אין סוף, filled with the emanation thereof, and is dominated by חכמה.

2. בריאה (*Beriyah*, "creation"): This is the first level of the creation of the physical Uni-

verse, but still lacking in particulars. It is dominated by בינה.

3. יצירה (*Yetzirah*, "formation"): This is the second level of the physical, in which creations assume particular forms. (You might call this the *hylomorphic* world.) It is dominated by חסד, גבורה, תפארת, נצח, הוד, and יסוד, the fabric of emotional experience.

4. עשיה (*Asiyah*, "making" / "doing"): This is the spiritual aspect of the world as we see it, fully formed. It is dominated by מלכות.

The kabbalistic approach to creation concerns the role of the ספירות in sustaining the physical world, as well as the origin of this very pattern.

Howard Smith makes a valiant effort to associate the four עולמות with stages of the physical unfolding of the Universe. He suggests that עצילות corresponds with the Universe "before the Big Bang." Here he really means before the origin of time,[21] which is distinct from our usage of "Big Bang." Then בריאה corresponds to the earliest moments of the physical Universe, through the end of inflation, but prior to the first existence of matter particles. This first existence of matter—reheating, as we have defined it in section 5—corresponds to the first formless matter, the Greek *hyle.* The period from the appearance of matter particles to the formation of the first atoms is the first appearance of form within matter, *hylomorphism,* and thus corresponds to יצירה. From the first atoms to the fully formed and differentiated Universe, the active process is not creation but reassembly. This is the period of עשיה.

Despite this appealing correspondence, I want to make clear that the mystical literature is fundamentally difficult for an empiricist. It often speaks in magical or ecstatic language. For example, consider the opening of one of the most foundational mystical accounts of creation, the *Sefer Ha-Yetzirah* / ספר היצירה, traditionally attributed to Abraham:

בשלשים ושתים נתיבות פליאות חכמה חקק יה ה׳ צבאות
אלקי ישראל אלקים חיים ומלך עולם אל שדי רחום וחנון
רם ונשא שוכן עד מרום וקדוש שמו וברא את עולמו
בשלשה ספרים בספר וספר וספור: עשר ספירות בלי
מה ועשרים ושתים אותיות יסוד שלש אמות ושבע כפולות
ושתים עשרה פשוטות: עשר ספירות בלימה מספר עשר
אצבעות חמש כנגד חמש וברית יחיד מכוונת באמצע
במלת הלשון וכמילת מעור: עשר ספירות בלימה

With thirty-two wondrous paths of wisdom, Y"H YHW"H, God of Israel, Living God and King of the World, El Shaddai, Merciful and Compassionate, High and Lofty One Dwelling Eternally Above, Holy is His Name [Isaiah 57:15]—[God] engraved and created [God's] world with three [instances of the root] ס-פ-ר (*s-p-r*): with סְפֹר (*s'por,* enumeration), סֵפֶר (*seifer,* scroll), and סִפּוּר (*sipur,* telling). Ten Sefirot-Without-What and twenty-two letters of foundation: Three mothers, seven pairs, and twelve simple ones. Ten Sefirot-Without-What: The number of ten fingers, five against five, covenant of the One fixed in the middle, like a word of the tongue or circumcision of the genitals. Ten are the numbers of the ineffable sefirot.

It is extremely challenging to imagine putting such language in any meaningful correspondence with the scientific understanding of the Universe. However, this is not altogether surprising. At this point, it is important to recognize that although we started off by distinguishing

[21] Howard, if you're reading this, I'd be happy to stop by CfA and discuss!

between cosmology and cosmogony, virtually everything in the actual text of בראשית is *cosmology.* For all we talk about creation *ex nihilo,* a plain reading of the text (the opening chapter, anyway) does not suggest creation from nothing. It only discusses creation from a starting point of primordial chaos—not *nothing.* Mysticism, on the other hand, is largely concerned with cosmogony. The great question of creation in this framework regards the transition from a state of nonbeing, "אין," to a state of being, "יש," in which *something* exists. We cannot expect a theory of cosmogony to align with a theory of cosmology.

Thus, I would suggest that kabbalistic interpretations of creation are nothing more or less than the traditional approach to expanding the collective ontology that I invoked in the previous section. We do this as scientists by developing new measurements and models, which give rise to new sorts of entities. This is not an option that is available to nonempirical theology. Rather, the mystical, the search for ecstatic, experiential reception of the divine, is the primary tool by which the concepts of existence can be expanded.

16 Tripartite dialectical interpretation

Let us now consider another interpretive framework for the creation narrative in light of physical cosmology, this time rooted in the very tensions that we have previously sought to ameliorate. In short, this framework recognizes that there are actually *two* competing creation narratives in the text of בראשית, specifically regarding the creation of humans. It then likens cosmology to a *third* version of the creation story, and suggests widening the conversation between the two textual narratives to include the third, forming a tripartite dialectic.

Let's begin by recalling the two stories of human creation. First, in the narrative of the six days, humans are created, male and female, in the image of God (1:27):

וַיִּבְרָא אֱלֹקִים | אֶת־הָאָדָם בְּצַלְמוֹ בְּצֶלֶם אֱלֹקִים בָּרָא אֹתוֹ
זָכָר וּנְקֵבָה בָּרָא אֹתָם׃

So God created Mankind in His own image, in the image of God He created him; male and female He created them.

Shortly afterwards, the creation of humanity is retold in slightly different form (2:7):

וַיִּיצֶר יְ׳ אֱלֹקִים אֶת־הָאָדָם עָפָר מִן־הָאֲדָמָה וַיִּפַּח בְּאַפָּיו
נִשְׁמַת חַיִּים וַיְהִי הָאָדָם לְנֶפֶשׁ חַיָּה׃ [...] וַיַּפֵּל יְ׳ אֱלֹקִים |
תַּרְדֵּמָה עַל־הָאָדָם וַיִּישָׁן וַיִּקַּח אַחַת מִצַּלְעֹתָיו וַיִּסְגֹּר בָּשָׂר
תַּחְתֶּנָּה׃ וַיִּבֶן יְ׳ אֱלֹקִים | אֶת־הַצֵּלָע אֲשֶׁר־לָקַח מִן־הָאָדָם
לְאִשָּׁה וַיְבִאֶהָ אֶל־הָאָדָם׃

And the Lord God formed man of the dust of the ground, and breathed into his nostrils the breath of life; and man became a living soul. [...] And the Lord God caused a deep sleep to fall upon the man, and he slept: and He took one of his sides, and closed up the flesh in its place, and of the side, which the Lord God had taken from the man, He made a woman, and brought her to the man.

The commentators take notice of this repetition. בראשית רבה identifies this as descriptive of two different creations, and interpret these creations as supplying different parts of a human:

שְׁתֵּי יְצִירוֹת, יְצִירָה מִן הַתַּחְתּוֹנִים וִיצִירָה מִן הָעֶלְיוֹנִים.

There were two acts of formation [involved in the creation of man], a formation from the lower realms and a formation from the upper realms.

An alternate interpretation is provided in terms of a human's natural inclinations:

שְׁנֵי יְצָרִים, יֵצֶר טוֹב וְיֵצֶר הָרָע.

[The two acts of formation (yetzirah), allude to the creation of man's] two inclinations (yetzarim), the good inclination and the evil inclination.

This begins a long tradition of commentators extracting meaning from the comparison of the two creations.

I find this remarkable, because that exactly what we have been doing in comparing the textual narrative to physical cosmology. These are two categories of creation story, and we should analyze the tensions between them exactly as we already do with the intratextual tensions. Here, part of the work has even been done for us by Joseph Soloveitchik. Soloveitchik was deeply concerned with the role of a spiritual person in an increasingly technical society. In his essay *The Lonely Man of Faith,* Soloveitchik used the tension between the two creations

as the basis for a theory of two kinds of "Adam" still with us today. Of the first Adam, created in the image of God, he writes:

> Adam the first is interested in just a single aspect of reality and asks one question only — "How does the cosmos function at all?" nor is he interested in the question, "What is its essence?" He is only curious to know how it works. […] Adam the first wants to be human, to discover his identity which is bound up with his humanity. […] His motto is success, triumph over the cosmic forces. […] Adam the first is the mathematical scientist who whisks us away from the array of tangible things […] into a formal relational world of thought constructs…

Of the second Adam, created from dust and breath, he writes:

> Adam the second is, like Adam the first, also intrigued by the cosmos. […] His inquiry is of a metaphysical nature and a threefold one. He wants to know: "Why is it?" "What is it?" "Who is it?" […] He wants to understand the living, "given" world into which he has been cast. Therefore, he does not mathematize phenomena or conceptualize things. He encounters the universe in all its colorfulness, splendor, and grandeur, and studies it with the naivete, awe and admiration of the child who seeks the unusual and the wonderful in every ordinary thing and event.

Soloveitchik's essay confronts the problem of how to unite these two different Adams. In particular, the tension in the textual description of their creation stays with us today in the tension between different parts of our own being. In this reading, the presentation of two different acts of creation is not the result of faulty integration of different sources, but rather a parable that gives us a framework for confronting the disharmony of our own being. The conflict is *deliberate:*

> The Biblical dialectic stems from the fact that Adam the first, majestic man of dominion and success, and Adam the second, the lonely man of faith, obedience, and defeat, are not two different people locked in an external confrontation as an "I" opposite a "thou", but one person who is involved in self-confrontation. […] Since the dialectical role has been assigned to man by God, it is God who wants the man of faith to oscillate between the faith community and the community of majesty, between being confronted by God in the cosmos and the intimate, immediate apprehension of God through the covenant, and who therefore willed that complete human redemption be unattainable.

I now invite you to additionally consider *Adam the third:* the human product of evolution in a mechanical Universe. Unlike his brothers Adam I and Adam II, Adam III was not created *ex nihilo* by a thoughtful creator. He was not designed. He arose as by undirected chance, as the outcome of physical and biological processes acting on stardust that fell upon solid rock. Adam III is animalistic, reductive, and antiteleological.

How does Adam III enter this dialectic? Clearly, Adam III is fundamentally incompatible with Adams I and II. He is not even describable in the revelatory language of תנ״ך. He is dispossessed of a creator. In contrast to Adam I, who is defined by his method of inquisition, Adam III is defined by his status in relation to the rest of the Universe. In particular, he has discovered again and again that he is not in a uniquely privileged position in the world. Adam I discovered that the Earth is not the center of the Universe, but it is Adam III who was *displaced* by the Copernican principle. If Adam I is the mathematical man, Adam III in

our present intellectual era is the *statistical* man. He is subject to the ebb and flow of pure chance—and this is another feature that makes him indescribable in the narrative. In the verbiage of section 14, he is man as defined in ontological categories of his own making.

If we read the two creations as providing context for our eventual self-confrontation, we cannot help but read the the physical story of creation as a deliberate provision of additional context for the same conflict. In particular, when considering בראשית as a religious text, this includes the premise that the author is aware of our future scientific accomplishments. Abraham Isaac Kook, in *Igrot HaRe'aya* 91, makes precisely this argument:

> The Torah related the story of creation in a closed manner and spoke in hints and parables. We know that the actual story behind creation is one of the Torah's deepest secrets (see Chagiga 11b). […] The main thing is the "knowledge of Hashem" that comes out of the episode and learning true morality. […] Similarly that which is uncovered over thousands of years about the manner of the world's development, which increases our knowledge, is still revealed to us cryptically. The main point is that there is a time and place for each matter, without Hashem leaving anything to chance.

In this paradigm, revelation is an ongoing process in which Adam III plays a central role. As we learn more about the physical Universe, we slowly and painfully fill in the picture of who we really are, with great conflict between the different parts of our humanity. My point here is that this is a *feature* of the narrative. In this reading, we are *supposed* to confront our split nature. The narrative reflects our primitive, premathematical self-conception. It was easy to see ourselves as purely Adam I and Adam II in the context of Eden. But the story of creation presages development of the human place in the world, just as we leave the garden. We cannot complete this maturation without developing our empirical understanding of the Universe we inhabit.

17 Reconstruction: cosmological אגדה on בראשית א׳

At this point, having discussed several interpretive frameworks that use physical cosmology to understand בראשית, I want to point out that we have been constantly darting around one idea: that if *you* were to write the textual narrative yourself, with modern knowledge of cosmology, you would have to make some very difficult choices. This suggests that a final interpretive lens can be achieved by literally making the attempt. If we try to reconstruct בראשית on the basis of our cosmological understanding, what do we get, and why?

This form of interpretation is not new at all. It is essentially aggadic midrash. Here, the midrashist explicates some mysterious passage by creating new narrative (**אגדה** / "aggadah") to fill the gaps in our understanding. We can attempt to do the same thing with the creation narrative, and rewrite it in midrashic form. Of course, to do this properly, we need to end up with a reconstructed text that corresponds in form to the original, working within the same structural constraints of length, meter, language, and allegory.

I attempted this exercise for the first three days of creation. Here is what I came up with—my midrash of cosmological creation.

בְּרֵאשִׁית הַזְּמַן הַיָקוּם הָיָה רֵיק מֵהַמִשָעָר וְאֵינֵנוּ נִתַן לְהַכִּיר בְּצוּרָה וְלֹא עָבַר רֶגַע שֶׁל זְמַן וְלֹא הָיְתָה מִדָּה שֶׁל מָקוֹם מֻבְדָּלֶת: וַיֹּאמֶר אֱלֹהִים יְהִי הַזְּמַן מִתְפַּתֵּחַ וְהַיָקוּם יִתְרַחֵב עַד שֶׁלֹּא יֵרָאֶה קָצֵהוּ וַיְהִי בֵן וַיְמָלֵא הַיָקוּם בְּאֵשׁ כְּבֵדָה בְּלִי גוּף: וַיְהִי עֶרֶב וַיְהִי בֹּקֶר יוֹם אֶחָד:

[1]At the beginning of time, the Universe was devoid of extension and unknowable in form, and no moment of time passed, nor was a measure of space distinct. [2]And God said, Let time move forward, and let the Universe grow so great that its edge cannot be seen; and it was so; and the Universe was filled with a thick fire devoid of corporeality. [3] And it was evening, and it was morning, the first day.

וַיֹּאמֶר אֱלֹהִים תִּקָּבֵץ הָאֵשׁ לְאוֹר וּמַיִם וְהָאֵשׁ תִּתְקָרֵר בֵּין הַמַּיִם וַתִּתְקָרֵר הָאֵשׁ לְאָבָק וָאֵפֶר וַיְהִי עֲנָן גָּדוֹל לִפְנֵי הָאוֹר: וַיַּרְא אֱלֹהִים כִּי הָאוֹר נִכְלָא בָּעֲנָן וַיֹּאמֶר אֱלֹהִים יִקָּבְצוּ הָאָבָק וְהָאֵפֶר וְיִקָּבְצוּ הָאָבָק וְהָאֵפֶר וַיִרְדֵּן הֶעָנָן וַיְמָלֵא הַיָקוּם אוֹר וַיַּרְא אֱלֹהִים כִּי טוֹב: וַיְהִי עֶרֶב וַיְהִי בֹּקֶר יוֹם שֵׁנִי:

[4]And God said, Let the fire form into light and water, and let the fire cool amid the water; and the fire cooled into dust and ashes, and they were a great fog before the light. [5]And God saw that the light was imprisoned by the fog, and God said, Let the dust and ash join together; and the dust and ash joined, and the fog receded, and the Universe was filled with light; and God saw that it was good. And it was evening, and it was morning, the second day.

וַיֵרֶךְ אוֹר הָאֵשׁ וְלֹא הָיָה מְאוֹר בַּשָׁמַיִם: וַיֹּאמֶר אֱלֹהִים יִמָּלֵא הַמַּיִם גַּלִּים גְּדוֹלִים וַיְהִי כֵן וַיִרְבּוּ הַגַּלִּים בַּמַּיִם וַיִתְנַפְּצוּ יַחַד: וַיֹּאמֶר אֱלֹהִים יִהְיוּ מְאוֹרוֹת בְּרֹאשֵׁי הַגַּלִּים וַיִמָּלְאוּ הַגַּלִּים מְאוֹרוֹת וַיְמַלֵּא הַיָקוּם אוֹר: וַיּוֹצִיאוּ הַמְאוֹרוֹת אֵפֶר כָּבֵד מִבְּעָרָתָם וַיִפְּלוּ לִבְרִים עַל הַמַּיִם וַיַּרְא אֱלֹהִים כִּי טוֹב: וַיִקְרָא אֱלֹהִים לִבְרִים אֶרֶץ וְלַמַּיִם קָרָא שָׁמַיִם: וַיְהִי עֶרֶב וַיְהִי בֹּקֶר יוֹם שְׁלִישִׁי:

[6]And the light of the fire grew dim, and there was no luminary in the sky. [7]And God said, Let the water be filled with great waves; and it was so; and the waves in the water grew larger and crashed together. [8]And God said, let there be luminaries at the tops of the waves; and the waves were filled with luminaries, and they filled the Universe with light. [9]And the luminaries issued forth heavy ashes from their burning, and they fell into islands upon the waters; and God saw that it was good. [10]And God called the islands "land," and the waters he called "sky." [11]And it was evening, and it was morning, the third day.

You can probably tell that this is more "scientifically accurate" than the original text, in the sense that it is easier to construct a correspondence between this text and our scientific chronology. This is in part because of chronological considerations—this is ordered as we know cosmic history unfolded—and in part because of ontological considerations. We have started from the physical ontology and projected it onto the narrative ontology rather than the other way around, and so it is much easier to identify elements of this midrash with entities of the physical ontology.

Still, this ontological projection was quite challenging, and a great deal of the meaning was lost in the process. Even some of the concepts I chose as simplifications of scientific notions might not have any meaning to the ancient audience. The relations between them are even less clear. It is thus doubtful that the resulting narrative conveys anything that would be interpreted as physically meaningful by the reader. At the same time, other aspects of meaning in the original text have also been lost, and that is what I want to focus on here.

The original text is "pragmatic," not in the sense that it provides anything of specific use to the reader, but in that establishes a scriptural context for objects in the reader's life. The discussion largely revolves around things that humans can *use*—from land and water to plants and animals to the Sun and Moon. Those things which are of no use are described by hardly more than that label alone. After all, "useless matter" wouldn't be a terrible interpretation of תהו ובהו, which is essentially the only useless object of the narrative. Moreover, beyond the physical objects of pragmatism, the whole of the entire creation story provides one enormously significant abstract object: the Jewish architecture of time. This story provides the basis for the structure of the week, with six days of work and one day of rest. In fact, the contents of the story are also deeply concerned with human measurement of time. The Sun and Moon are created not to sustain plants, but to mark days, months, and years.

The question we must ask whenever we interpret the narrative in a nonliteral fashion is: why? If the story is not literal, *what is it for?* What is it doing instead of telling us a plain story about the Universe? A traditional viewpoint is characterized by the writing of Samson Raphael Hirsch, a 19th century commentator:

> […] Creation was from nothing, a truth that forms the foundation of the consciousness that God's teachings seek to build for us. The opposite, the eternity of the material of the world, which only allows the Creator to be the creative designer and which also forms the foundation of pagan consciousness to this day, is not only a metaphysical lie that robs people's cosmogonic ideas of truth, i.e., their agreement with reality: it is the even more pernicious denial of all freedom in God and in man, which undermines all morality. If the world creator had been given the material, he could not create the absolutely good world from the given material, but only the relatively best world.
>
> Es ist damit die Schöpfung aus Nichts […] eine Wahrheit, die den Grundstein des Bewusstseins bildet, welches die Lehre Gottes uns aufbauen will. Das Gegenteil, die Urewigkeit des Weltstoffes, die den Schöpfer nur als den gestaltenden Bildner zulässt und die ebenso den Grundstein des heidnischen Bewusstseins bis auf den heutigen Tag bildet, ist nicht nur eine metaphysische Lüge, die den kosmogonischen VorStellungen der Menschen die Wahrheit, d. h. die Übereinstimmung mit der Wirklichkeit geraubt: sie ist die noch weit verderblichere, alle Sittlichkeit untergrabende Leugnung aller Freiheit in Gott und im Menschen. War dem Weltbildner der Stoff gegeben, so konnte er aus dem gegebenen Stoff nicht die absolut gute, sondern nur die relativ beste Welt gestalten.

In short, for Rav Hirsch, the essential feature of the creation narrative is simply that creation takes place *ex nihilo,* regardless of the actual sequence of that creation or the specific actions of that creator. As he points out, this is a cosmo*gonic* statement rather than a cosmo*logical* statement. Does my midrash achieve this feature? I would argue that it does, at least as well as the original text, which is really mostly cosmological and barely cosmogonic. What, then, do we make of the significance of the rest of the story departing from literalism?

The present exercise in אגרה suggests that a plain story produces a very different narrative with virtually no attachment to the life of the reader. The strength of the original narrative lies not in poetry, but in practicality. This chapter provides the basis to assign ritual meaning to various items in the physical world, including plants, animals, and periods of time. For a document such as the Torah, this might be the highest aspiration.

18 Final thoughts

I want to close with a few parting thoughts. The preceding sections have of course had a very different character from that of part I. There, I was firmly in my comfort zone, and I was simply telling you a story that is fantastically well supported by empirical evidence. Part II is largely my unsubstantiated opinion in a framework of textual analysis that strongly deviates from my training and my day job. Still, there are a few arguments from this exercise that I think are interesting, and that I want to highlight.

1. The conflict between narrative and scientific conceptions of the Universe is reactionary, and any insistence on strict literalism is not just counterproductive but also countertraditional. We have comfortably employed metaphor in the interpretation of the creation narrative for thousands of years without worrying about falsifying the narrative as though it were a scientific theory. This only seems to resurface at cultural moments when theologians act as though under threat.

2. There are certain concepts of creation that lack a mode of expression in the linguistic context of the narrative. These ideas can only be formulated and expressed as our collective ontology develops additional capacities. It is thus impossible for the narrative to tell a complete and true story of creation on its own.

3. The creation narrative itself is split into two stories in tension, and this is an asset to understanding its relation to physical cosmology. If we interpret the physical chronology as a third story of creation, in tension with each of the other two, we can interpret the narrative as providing a framework for the dialectic of our collective maturation.

4. The creation narrative is eminently pragmatic. If we attempt to rewrite it on the basis of the scientific understanding of cosmology, we not only lose most of the scientific meaning, but we lose a great deal of ritual context that is established by the narrative.

With these points in mind, I want to suggest that our improving understanding of cosmology does not always simplify our understanding of our existence. There are some philosophical puzzles which are *resolved* by physical cosmology, but there are others that are deepened. Both outcomes are valuable.

More generally, I point out that the boundary between science and philosophy has long been fuzzy and inconsistent. Aristotle's philosophy of cosmology was treated in much the same way that we would treat scientific statements today—not in the sense that it was examined critically and quickly falsified, but rather in the sense that it was regarded as a reliable understanding of the Universe constructed through reason. After all, until fairly recently, scientists were called *natural philosophers*. Of course, a great deal of philosophical effort has gone into securing the boundaries of empirical scientific work, with the definitions of the scientific method and falsifiability, for example, only emerging recently.

But still, the barrier between philosophy and science is doubtless permeable. This is true in both directions. In one, we carry nonempirical ontological notions into scientific work

that, while separate from the predictive mechanisms of that work, nonetheless inform our conceptualization of the Universe. In the other, we develop new methods of inquiry that put formerly philosophical questions into the realm of empirical study.

Personally, in the context of the encounter between physical cosmology and בראשית I am most intrigued by puzzle of entropy and time. As I argued in section 14, the statistical notion of an arrow of time makes it possible for us to envision a "creation" of time without inconsistency. But this picture yields a counterintuitive corollary. The narrative of creation is one that moves from disorder to order—at least, that has been the common conception. But if we loosely interpret "entropy" as "disorder," creation must ultimately proceed from order to disorder.[22] How can this be? The entropic arrow of time suggests a very different reading: the Universe must begin in an ordered state and move towards disorder *as an intrinsic feature of the passage of time.* In other words, the creation of time itself, which we might loosely interpret as part of "ordering" the Universe, is reliant upon increasing "disorder" in the form of entropy, which is itself reliant upon the creation of state that does not maximize entropy. This shifts our concept of the creation narrative: in some sense, in order that the story should happen at all, God creates the possibility for increasing disorder.

Is this some deep realization about the nature of the creation narrative? Maybe. I don't know. There are many reasons why I don't know, but one reason is most interesting to me: the above statement is made on the basis of our present collective ontology. As this ontology continues to expand, I don't know what role these notions will take in the distant future. Perhaps one day we will discover a more fundamental notion that underpins the behavior of statistical ensembles in the physical Universe, and perhaps that will change how we understand the arrow of time, and the above argument will be rendered obsolete. That, to me, is the most exciting possibility. By expanding our effective vocabulary for describing the world, we open ourselves up to new interpretations of the creation narrative that we could never have conceptualized otherwise.

In all of this, I think any serious student of Jewish theology must recognize, as did Rav Kook, that a revelatory Torah could not be a divine revelation without the revelator expecting us to ask these very questions. Our development of physical cosmology is meant to be *part* of this story. In other words, תורה היא, וללמוד אנחנו צריכים—this is Torah, and we must learn it.

Acknowledgments

I thank my co-teacher Jeff Cohen for providing the motivation for me to take the time for this exercise. I thank Tamar Wenocur for valuable contributions to section 17, and for making me aware of the work of Gerald Schroeder. Many of the Jewish sources that have been copied here were obtained from Sefaria (www.sefaria.org). Some citations have yet to be added in this version. I will update this document with appropriate references as soon as possible.

[22] It is so important to take extreme care in considering "entropy" to be something similar to "disorder." The correspondence is imprecise, and modern-day young Earth creationists have made much of misinterpreting the nature of entropy. So, please don't read too far into this analogy.

References

[1] **BOSS** Collaboration, S. Alam et al., *The Eleventh and Twelfth Data Releases of the Sloan Digital Sky Survey: Final Data from SDSS-III*, *Astrophys. J. Suppl.* **219** (2015), no. 1 12, [arXiv:1501.00963].

[2] **WMAP** Collaboration, C. L. Bennett et al., *Nine-Year Wilkinson Microwave Anisotropy Probe (WMAP) Observations: Final Maps and Results*, *Astrophys. J. Suppl.* **208** (2013) 20, [arXiv:1212.5225].